

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Syngas compressor Antisurge cooler-1-AK.pvdb

Date of Analysis : Sep 7,2026 1:18pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1605.000 mm. ID
 Vessel Design Length, Tangent to Tangent 8523.97 mm.
 Specified Datum Line Distance 0.00 mm.
 Shell Side Design Temperature 200 °C
 Channel Side Design Temperature 85 °C
 Shell Side Design Pressure 191.000 bars
 Channel Side Design Pressure 147.000 bars
 Shop Shell Side Test Pressure 249.052 bars
 Field Shell Side Test Pressure 255.203 bars
 Shop Channel Side Test Pressure 191.540 bars
 Field Channel Side Test Pressure 197.942 bars
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	+IT
Nozzle	SA-266 2	...	...	K03506	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	+IT
Flg Bolting	SA-193 B7	...	64 < t <= 100	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Blind Flg	147.332	0.00	151.000	3.0000	N/A	No
Ch Flg	147.157	0.00	147.000	3.0000	N/A	No
LHS Ch Shell	147.158	0.00	147.000	3.0000	N/A	No
Main Shell	191.008	0.00	192.000	3.0000	N/A	No
Dish End	191.008	0.00	195.000	3.0000	No	No

Liquid Level: 3390.00 mm. Dens.: 0.000 kg./cm³ Sp. Gr.: 0.050

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Body Flg	608.2	605.0	613.0	605.0	597.6	...	1.00	1.00
Body Flg	1383.2	775.0	598.0	585.0	581.1	...	1.00	1.00
Cylinder	2583.2	1200.0	95.0	95.0	94.8	...	1.00	1.00
Cylinder	8473.2	5600.0	125.0	125.0	124.6	...	1.00	1.00
Ellipse	8524.0	50.8	130.0	118.0	115.5	...	1.00	1.00



Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
T2-Tube outlet	WRC-297	0.986	Shell	Passed
T1-Tube Inlet	WRC-297	0.986	Shell	Passed
S1-Shell Inlet	WRC-297	0.986	Shell	Passed
S2-Shell outle	WRC-297	0.986	Shell	Passed

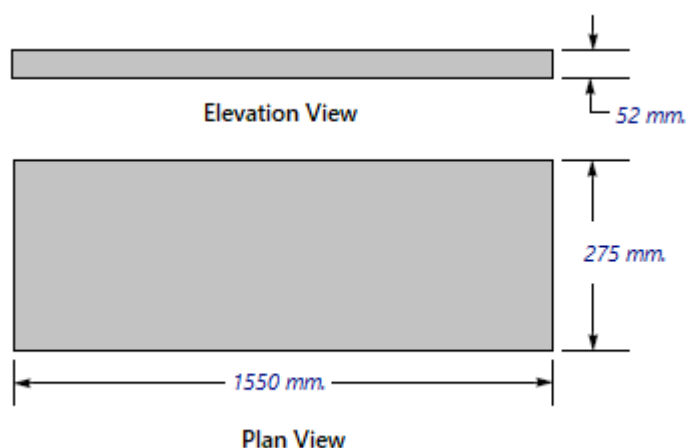
Foundation/Support Summary

Saddle Parameters:

Saddle Width	250.000 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1300.000 mm.
Wear Pad Width	765.000 mm.
Wear Pad Thickness	20.000 mm.
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	755.000 mm.

Baseplate Length	1550.000 mm.
Baseplate Thickness	52.000 mm.
Baseplate Width	275.000 mm.
Number of Ribs (including outside ribs)	5
Rib Thickness	20.000 mm.
Web Thickness	20.000 mm.
Height of Center Web	500.000 mm.
Number of Bolts in Baseplate	4

Baseplate Sketch



Maximum Tensile Bolt Load 24. Kg

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	209988.22 Kg
Maximum Transverse Saddle Shear Load	7574.08 Kg
Maximum Longitudinal Saddle Shear Load	15148.15 Kg

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	27420.06 Kg
Maximum Transverse Saddle Shear Load	393.50 Kg
Maximum Longitudinal Saddle Shear Load	228.64 Kg

Weights:

Fabricated - Bare w/o removable internals	150515.8 kg.
Shop Test - Fabricated + water (full)	164998.1 kg.
Shipping - Fab. + rem. intls.+ shipping app.	150882.5 kg.



Erected	- Fab. + rem. intls.+ insul. (etc)	150882.5	kg.
Empty	- Fab. + intls. + details + wghts.	150882.5	kg.
Operating	- Empty + operating liquid (no ca)	157279.7	kg.
Field Test	- Empty weight + test medium (full)	164496.2	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
T2-Tube outlet	147	...	...	OK	21.86	OK	Passed
T1-Tube Inlet	147	...	...	OK	21.88	OK	Passed
S1-Shell Inlet	192	...	...	OK	31.97	OK	Passed
S2-Shell outlet	192	...	...	OK	31.97	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 192 Nozzle : S1-Shell Inlet [Shellside]
 Minimum MAWP Nozzles : 147 Nozzle : T2-Tube outlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	T2-Tube outlet	1983.175	90.000	662.850
30	T1-Tube Inlet	1983.175	270.000	662.850
40	S1-Shell Inlet	4373.174	90.000	767.701
40	S2-Shell outlet	6657.894	270.000	767.701

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
T2-Tube outlet	16.000 in	160	RTJWn	16.000	40.488	...	...	458.43	1500
T1-Tube Inlet	16.000 in	160	RTJWn	16.000	40.488	...	...	458.43	1500
S1-Shell Inlet	18.000 in	140	RTJWn	18.000	39.675	...	...	514.83	1500
S2-Shell outlet	18.000 in	140	RTJWn	18.000	39.675	...	...	514.83	1500

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
T2-Tube out	SA-266 2	90.000	10.000	...	...	...
T1-Tube Inl	SA-266 2	90.000	10.000	...	...	...
S1-Shell In	SA-266 2	118.000	10.000	...	...	...
S2-Shell ou	SA-266 2	118.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
T2-Tube outlet	1983.175	90.0	300.00	0.00	LHS Ch Shell
T1-Tube Inlet	1983.175	270.0	300.00	0.00	LHS Ch Shell
S1-Shell Inlet	4373.174	90.0	300.00	0.00	Main Shell
S2-Shell outlet	6657.895	270.0	300.00	0.00	Main Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 613.0mm. THK X 3390.0mm. OD	SA-266 2
1	WELD NECK FLANGE: 598.0mm. THK X 3390.0mm. OD	SA-266 2
1	CYLINDER: 95.0mm. THK X 1605.0mm. ID X 1200.0mm.	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 130.0mm. THK X 1605.0mm. ID X 50.8mm.	SA-516 70
7	BAFFLES: 1450.0mm DIA X 8.0mm THK	SA-516 70
12	TIE RODS: 25mm DIA X 5000mm	SA-193 B7M
1	TUBESHEET: 1857.0mm X 290.0mm THK	SA-266 2
1126	TUBES: 5150.0mm X 25.4mm DIA X 3.4mm THK	SA-179
2	SADDLE: 250.0mm X 120 DEG	SA-516 70
1	INSULATION: 5600mm X 50mm THK	Hot Insulation
1	INSULATION: 452mm X 50mm THK	
1	GASKET: 1972.0mm. OD X 1637.0mm. ID	...
52	BODY FLANGE BOLTS: 95.2mm. DIA	SA-193 B7
104	NUTS FOR BODY FLANGE BOLTS: 95.2mm. DIA	...
1	CYLINDRICAL SEGMENT 125.0mm. THK X 3239.7mm. X 5600.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 125.0mm. THK X 1802.5mm. X 5600.0mm.	SA-516 70
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Main Shell	[8]	!	-10	-10		1.000	125.000	1.00	Yes
S1-Shell Inlet	[1]	B	28	23		0.913	34.715	1.00	Yes
Nozzle Flg	[5]	A	-18	-32					
S2-Shell outle	[1]	B	28	23		0.914	34.715	1.00	Yes
Nozzle Flg	[5]	A	-18	-32					
Tubesheet: SS	[13]	!	-10	-12		0.962	125.000	1.00	No
Warmest MDMT:			28	23					
Blind Flg	[11]	!	0	-2		0.966	151.250	1.00	No
Ch Flg	[11]	!	0	-2		0.966	95.000	1.00	Yes
LHS Ch Shell	[8]	!	0	0		0.999	95.000	1.00	Yes
T2-Tube outlet	[1]	B	30	30		0.996	95.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
T1-Tube Inlet	[1]	B	30	30		0.998	95.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
Tubesheet: CS	[14]	!	0	-17		0.694	95.000	1.00	No
Bolting	[21]		-40						
Warmest MDMT:			30	30					
Exchanger Side		Computed MDMT			Required MDMT		Pass/Fail		
Channel/Tube		30.0			0.0		Fail		

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	191	bars
Channel Side Design Pressure	147	bars
Shell Side Design Temperature	200.0	°C
Channel Side Design Temperature	85.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Sour	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-10.0	°C
MDMT (CET), Tube Side	0.0	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Field Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	0 mm.
Percent Wind for Hydrotect	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm.
Distance Upwind of Crest	Lh 0 mm.
Distance from Crest to the Vessel	x 0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
--	-----



Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Blind Flg
Distance "FROM" to "TO"	605 mm.
Flange Outside Diameter	3390 mm.
Element Thickness	605 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	613 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2 +IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	0 mm.
Yield Stress, Operating	229.57 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3390 mm.
Liquid Density	0.0009984 kg./cm ³

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Flange
Description	Ch Flg
Distance "FROM" to "TO"	775 mm.
Flange Inside Diameter	1605 mm.
Element Thickness	585 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	598 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C



Effective Diameter Multiplier 1.2
 Material Name SA-266 2 +IT
 Perform Flange Stress Calculation (Y/N) Y
 Weld is pre-Heated No

Element From Node 20
 Detail Type Liquid
 Detail ID Liquid: 20
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 1605 mm.
 Liquid Density 0.0009984 kg./cm³

Element# 3/5
 Element From Node 30
 Element To Node 40
 Element Type Cylinder
 Description LHS Ch Shell
 Distance "FROM" to "TO" 1200 mm.
 Inside Diameter 1605 mm.
 Element Thickness 95 mm.
 Internal Corrosion Allowance 3 mm.
 Nominal Thickness 95 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 147 bars
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Allowable Stress, Ambient 138 N./mm²
 Allowable Stress, Operating 138 N./mm²
 Allowable Stress, Hydrotest 234 N./mm²
 Material Density 0.00775 kg./cm³
 P Number Thickness 0 mm.
 Yield Stress, Operating 242 N./mm²
 UCS-66 Chart Curve Designation Impact Tested
 External Pressure Chart Name CS-2
 UNS Number K02700
 Product Form Plate
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 30
 Detail Type Liquid
 Detail ID Liquid: 30
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 1605 mm.
 Liquid Density 0.0009984 kg./cm³

Element From Node 30
 Detail Type Nozzle
 Detail ID T2-Tube outlet
 Dist. from "FROM" Node / Offset dist 600 mm.
 Nozzle Diameter 16 in.
 Nozzle Schedule 160
 Nozzle Class 1500
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 952.15 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2

Element From Node 30
 Detail Type Nozzle
 Detail ID T1-Tube Inlet
 Dist. from "FROM" Node / Offset dist 600 mm.
 Nozzle Diameter 16 in.



Nozzle Schedule	160	
Nozzle Class	1500	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1014.4	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element#	4/5
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	5600 mm.
Inside Diameter	1605 mm.
Element Thickness	125 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	125 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	200 °C
Design External Pressure	0 bars
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	755 mm.
Width of Saddle	250 mm.
Height of Saddle at Bottom	1300 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	765 mm.
Thickness of Wear Plate	20 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	40
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	4823.9 mm.
Width of Saddle	250 mm.
Height of Saddle at Bottom	1300 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	765 mm.
Thickness of Wear Plate	20 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	7.9999998e-002
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40



Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 1605 mm.
 Liquid Density 0.50120E-04 kg./cm³

Element From Node 40
 Detail Type Insulation
 Detail ID Hot Insulation
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Insulation 5600 mm.
 Thickness of Insulation 50 mm.
 Density 0.0001918 kg./cm³

Element From Node 40
 Detail Type Nozzle
 Detail ID S1-Shell Inlet
 Dist. from "FROM" Node / Offset dist 1500 mm.
 Nozzle Diameter 18 in.
 Nozzle Schedule 140
 Nozzle Class 1500
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 2586.4 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2

Element From Node 40
 Detail Type Nozzle
 Detail ID S2-Shell outlet
 Dist. from "FROM" Node / Offset dist 3784.7 mm.
 Nozzle Diameter 18 in.
 Nozzle Schedule 140
 Nozzle Class 1500
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 1564.5 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2

Element From Node 40
 Detail Type Weight
 Detail ID Additional weig
 Dist. from "FROM" Node / Offset dist 2000 mm.
 Miscellaneous Weight 10000 Kg
 Offset from Element Centerline 0 mm.

Element# 5/5
 Element From Node 50
 Element To Node 60
 Element Type Elliptical
 Description Dish End
 Distance "FROM" to "TO" 50.8 mm.
 Inside Diameter 1605 mm.
 Element Thickness 118 mm.
 Internal Corrosion Allowance 3 mm.
 Nominal Thickness 130 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 191 bars
 Design Temperature Internal Pressure 200 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 200 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No

Element From Node 50



Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1605 mm.
Liquid Density	0.50120E-04 kg./cm ³

Element From Node	50
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	452.05 mm.
Thickness of Insulation	50 mm.
Density	0.0001918 kg./cm ³



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Blind Flg		608.175	...	605	...
Ch Flg		1383.17	...	775	...
LHS Ch Shell		2583.18	...	1200	...
Main Shell		8473.17	...	5600	...
Dish End		8523.97	...	50.8	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flange Input Data Values Description: New Flange :

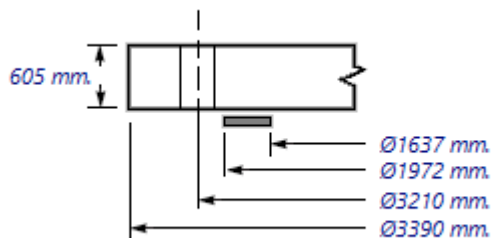
Blind Flg

Flange Type		Blind
Design Pressure	P	147.33 bars
Design Temperature		85 °C
Internal Corrosion Allowance	ci	3.0000 mm.
External Corrosion Allowance	ce	0.0000 mm.
Use Corrosion Allowance in Thickness Calcs.		Yes
Flange Outside Diameter	A	3390.000 mm.
Flange Thickness	t	605.0000 mm.
Flange Material		SA-266 2 +IT
Flange Material UNS number		K03506
Flange Allowable Stress At Temperature	Sfo	138.00 N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00 N./mm ²
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	159.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00 N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L	10084.513 mm.
Diameter of Bolt Circle	C	3210.000 mm.
Nominal Bolt Diameter	a	95.2500 mm.
Type of Threads	TEMA Thread Series	
Number of Bolts		52
Flange Face Outside Diameter	Fod	1983.000 mm.
Flange Face Inside Diameter	Fid	1605.000 mm.
Flange Facing Sketch		Sketch 1a
Gasket Outside Diameter	Go	1972.000 mm.
Gasket Inside Diameter	Gi	1637.000 mm.
Gasket Factor	m	2.0000
Gasket Design Seating Stress	y	17.24 N./mm ²
Gasket Thickness	tg	3.1750 mm.
Raised face depth	Fd	0.0000 mm.



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Gasket Contact Width,	$N = (G_o - G_i)/2$	167.500 mm.
Basic Gasket Width,	$b_o = N/2$	83.750 mm.
Effective Gasket Width,	$b = C_b * \sqrt{b_o}$	23.061 mm.
Gasket Reaction Diameter,	$G = G_o - 2 * b$	1925.878 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1925.8778^2 \cdot 147.332$$

$$= 4376650.000 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 23.0611 \cdot \pi \cdot 1925.8778 \cdot 2.0 \cdot 147.33$$

$$= 838517.812 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(4376650 + 838518 + 0, 0)$$

$$= 5215168.000 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 17.24 \cdot 23.0611 \cdot \pi \cdot 1925.878 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 245246.797 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((5215168 + 0 + 4 \cdot 0/1925.88)/159, 245247/159)$$

$$= 3216.679 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	3216.679	3391.064	
Radial Distance between Bolts and Edge:	88.900	90.000	
Circ. Spacing between the Bolts:	193.675	193.815	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 3391.064 \cdot 159.0 / (17.24 \cdot \pi (1972.0 + 1637.0))$$

$$= 275.876 \text{ mm. [Note: Exceeds actual gasket width, 167.500]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 159.0(3216.679 + 3391.064)/2$$

$$= 5356532.50 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 5215168.00 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (3210.0 - 1925.8778)/2$$

$$= 642.0611 \text{ mm.}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 5215168 \cdot 642.0611 \cdot 1.0 / (602.0^2 \cdot 1925.8778) + 0.3 \cdot 1.0 \cdot 147.33 \cdot 1925.8778^2 / 602.0^2$$

$$= 134.63 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 5356533 \cdot 642.0611 \cdot 1.0 / (602.0^2 \cdot 1925.8778)$$

$$= 91.82 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 5215168/3391.064$$



= 150.82 N./mm²

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)

= (245247/3391.064)

= 7.09 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	134.63	138.00	91.82 138.00
Bolting	150.82	159.00	7.09 159.00

Reqd. Blind Flange Thickness at Center 597.615 mm.

Reqd. Blind Flange Thickness at Gasket 494.046 mm.

Estimated M.A.W.P. (Operating) 151 bars

Estimated Finished Weight of Flange at given Thk. 42322.8 kg.

Estimated Unfinished Weight of Forging at given Thk 42322.8 kg.

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	133.30	138.00	90.91 138.00
Bolting	150.82	159.00	7.09 159.00

Minimum Req'd Thickness, ambient, no Corrosion: 594.639 mm.

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Flange Input Data Values **Description: Ch Flg** :

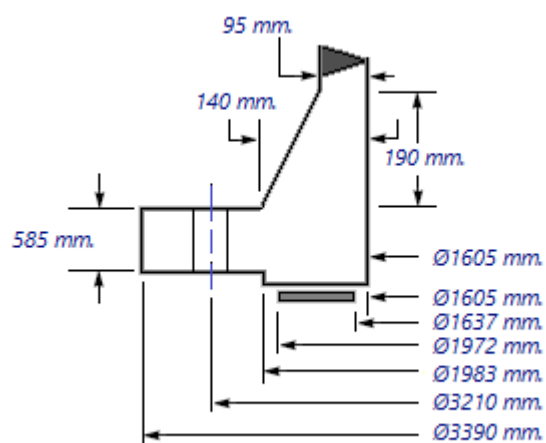
Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	147.16	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1605.000	mm.
Flange Outside Diameter	A	3390.000	mm.
Flange Thickness	t	585.0000	mm.
Thickness of Hub at Small End	go	95.0000	mm.
Thickness of Hub at Large End	gl	140.0000	mm.
Length of Hub	h	190.0000	mm.
Flange Material		SA-266 2 +IT	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	153.14	N./mm ²
Flange Allowable Stress At Ambient	Sfa	165.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	3210.000	mm.
Nominal Bolt Diameter	a	95.2500	mm.
Type of Threads	TEMA Thread Series		
Number of Bolts		52	
Flange Face Outside Diameter	Fod	1983.000	mm.
Flange Face Inside Diameter	Fid	1605.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1972.000	mm.
Gasket Inside Diameter	Gi	1637.000	mm.
Gasket Factor	m	2.0000	
Gasket Design Seating Stress	y	17.24	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (147.16 \cdot (1605.0/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 147.16) + Ca$$

$$= 94.7742 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 92.0) / (805.5 + 0.6 \cdot 92.0)$$

$$= 147.496 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 95.0) / (802.5 + 0.6 \cdot 95.0)$$

$$= 152.518 \text{ bars}$$

Corroded Flange Thickness, tc = T-ci	582.000 mm.
Corroded Flange ID, Bcor = B+2*ci	1611.000 mm.
Corroded Large Hub, g1Cor = g1-ci	137.000 mm.
Corroded Small Hub, g0Cor = go-ci	92.000 mm.
Code R Dimension, R = ((C-Bcor)/2)-g1cor	662.500 mm.
Gasket Contact Width, N = (Go - Gi)/2	167.500 mm.
Basic Gasket Width, bo = N/2	83.750 mm.
Effective Gasket Width, b = Cb*sqrt(bo)	23.061 mm.
Gasket Reaction Diameter, G = Go - 2 * b	1925.878 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1925.8778^2 \cdot 147.157$$

$$= 4371452.500 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 23.0611 \cdot \pi \cdot 1925.8778 \cdot 2.0 \cdot 147.16$$

$$= 837522.000 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1611.0^2 \cdot 147.1573$$

$$= 3058858.250 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 4371453 - 3058858$$

$$= 1312594.125 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(4371453 + 837522 + 0, 0)$$

$$= 5208974.500 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 17.24 \cdot 23.0611 \cdot \pi \cdot 1925.878 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 245246.797 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((5208975 + 0 + 4 \cdot 0 / 1925.88) / 159, 245247 / 159)$$

$$= 3212.859 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 95.25 + 6 \cdot 582.0 / (2.0 + 0.5)$$

$$= 1587.300 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 3210.0 \cdot \sin(\pi / 52)$$

$$= 193.815 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{193.815 / (2 \cdot 95.25 + 582.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	3212.859	3391.064	
Radial Distance between Hub and Bolts:	112.713	662.500	
Radial Distance between Bolts and Edge:	88.900	90.000	
Circ. Spacing between the Bolts:	193.675	193.815	1587.300

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 3391.064 \cdot 159.0 / (17.24 \cdot \pi(1972.0 + 1637.0))$$

$$= 275.876 \text{ mm. [Note: Exceeds actual gasket width, 167.500]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 159.0 \cdot 3391.064$$

$$= 5497897.00 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 5208975 - 4371453$$

$$= 837522.00 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (3210.0 - 1925.8778) / 2$$

$$= 642.0611 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((3210.0 - 1611.0) / 2 + 642.0611)$$

$$= 720.7805 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (3210.0 - 1611.0 - 137.0) / 2$$

$$= 730.9999 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	3058858.	730.9999	1.0000	2236070.
Face Pressure, Mt	1312594.	720.7805	1.0000	946111.
Gasket Load, Mg	837522.	642.0611	1.0000	537751.
Gasket Seating, Matm	5497897.	642.0611	1.0000	3530055.

Total Moment for Operation, Mo 3719931. Kg-m.
 Total Moment for Gasket seating, Mg 3530055. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 384.983 mm.
 Hub Ratio, h/h0 = HL / H0 0.494

Thickness Ratio, $g1/g0 = (g1Cor/goCor)$ 1.489

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.849
 Factor V per 4.16.5 0.316
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
 = 3390.0/1611.0
 = 2.104

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 = $(0.66845 + (5.7169(2.104^2 \cdot \log_{10}(2.104))) / (2.104^2 - 1)) / (2.104 - 1)$
 = 2.766

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 = $(2.104^2(1 + 8.55246 \cdot \log_{10}(2.104)) - 1) / ((1.0472 + 1.9448 \cdot 2.104^2) \cdot (2.104 - 1))$
 = 1.469

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
 = $(2.104^2(1 + 8.55246 \cdot \log_{10}(2.104)) - 1) / (1.36136(2.104^2 - 1) \cdot (2.104 - 1))$
 = 3.040

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
 = $(2.104^2 + 1) / (2.104^2 - 1)$
 = 1.583

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g0^2 \cdot h0 / V$
 = $3.04 \cdot 92.0^2 \cdot 384.983 / 0.316$
 = 31345048.000 mm.³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.849/384.983
 = 0.002 1/mm.

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / T + t^3/d$
 = $(582.0 \cdot 0.002 + 1) / 1.469 + 582.0^3/31345048$
 = 7.844

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g1^2)$
 = $(1.0 \cdot 3719931/1611.0) / (7.8437 \cdot 137.0^2)$
 = 153.81 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g1^2)$
 = $(1.0 \cdot 3530055/1611.0) / (7.8437 \cdot 137.0^2)$
 = 145.96 N./mm²

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 = $(2.7105 \cdot 3719931/1611.0) / (7.8437 \cdot 582.0^2)$
 = 23.10 N./mm²

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 = $(2.7105 \cdot 3530055/1611.0) / (7.8437 \cdot 582.0^2)$
 = 21.92 N./mm²

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned}
 &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO \\
 &= (2.766 \cdot 3719931 / (582.0^2 \cdot 1611.0)) - 1.5834 \cdot 23 \\
 &= 148.33 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned}
 &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA \\
 &= (2.766 \cdot 3530055 / (582.0^2 \cdot 1611.0)) - 1.5834 \cdot 22 \\
 &= 140.76 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned}
 &= (SHo + \max(SRo, STo)) / 2 \\
 &= (154 + \max(23, 148)) / 2 \\
 &= 151.07 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned}
 &= (SHA + \max(SRa, STa)) / 2 \\
 &= (146 + \max(22, 141)) / 2 \\
 &= 143.36 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= Wm1 / Ab \\
 &= 5208975 / 3391.064 \\
 &= 150.64 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (Wm2 / Ab) \\
 &= (245247 / 3391.064) \\
 &= 7.09 \text{ N./mm}^2
 \end{aligned}$$

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	153.81	229.71	145.96	247.50
Radial Flange	23.10	153.14	21.92	165.00
Tangential Flange	148.33	153.14	140.76	165.00
Maximum Average	151.07	153.14	143.36	165.00
Bolting	150.64	159.00	7.09	159.00

Minimum Required Flange Thickness 581.127 mm.

Estimated M.A.W.P. (Operating) 147 bars
 Estimated Finished Weight of Flange at given Thk. 32686.8 kg.
 Estimated Unfinished Weight of Forging at given Thk 42062.5 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$\begin{aligned}
 &= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki) \\
 &= 52.14 \cdot 3530055 / 1.0 \cdot 9806.637 \cdot 0.316 / (7.713 \cdot 201334 \cdot 92.0^2 \cdot 384.983 \cdot 0.3) \\
 &= 0.375 \text{ (should be } \leq 1)
 \end{aligned}$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$\begin{aligned}
 &= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki) \\
 &= 52.14 \cdot 3719931 / 1.0 \cdot 9806.637 \cdot 0.316 / (7.713 \cdot 197675 \cdot 92.0^2 \cdot 384.983 \cdot 0.3) \\
 &= 0.403 \text{ (should be } \leq 1)
 \end{aligned}$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$\begin{aligned}
 &= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki) \\
 &= 52.14 \cdot 3530055 / 1.0 \cdot 9806.637 \cdot 0.316 / (7.844 \cdot 201334 \cdot 92.0^2 \cdot 384.983 \cdot 0.3) \\
 &= 0.369 \text{ (should be } \leq 1)
 \end{aligned}$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$\begin{aligned}
 &= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki) \\
 &= 52.14 \cdot 3719931 / 1.0 \cdot 9806.637 \cdot 0.316 / (7.844 \cdot 197675 \cdot 92.0^2 \cdot 384.983 \cdot 0.3) \\
 &= 0.396 \text{ (should be } \leq 1)
 \end{aligned}$$



Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	152.92	247.50	144.87	247.50
Radial Flange	23.64	165.00	22.40	165.00
Tangential Flange	145.82	165.00	138.14	165.00
Maximum Average	149.37	165.00	141.51	165.00
Bolting	150.64	159.00	7.09	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 556.692 mm.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Blind Flg		147.33	613	3	3390	138	1
Ch Flg		147.16	598	3	1605	153.14	2
LHS Ch Shell		147.16	95	3	1605	138	3
Main Shell		191.01	125	3	1605	138	4
Dish End		191.01	130	3	1605	138	5

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Blind Flg		147	150.682	152.524	605	597.615	1
Ch Flg		147	147.339	152.518	585	581.127	2
LHS Ch Shell		147	147.339	152.518	95	94.7744	3
Main Shell		191	191.578	196.566	125	124.599	4
Dish End		191	195.175	199.96	118	115.506	5

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

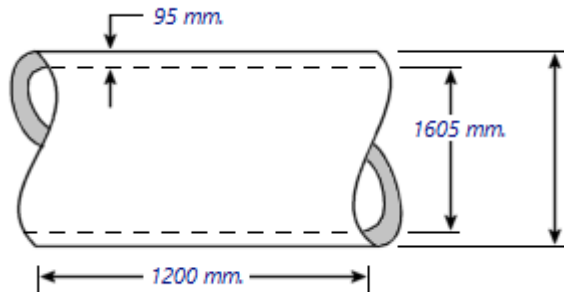
Shell Side MAWP = 192 bars
 Shell Side MAPnc = 197 bars
 Channel Side MAWP = 147 bars
 Channel Side MAPnc = 153 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	147.158 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1200.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1605.000 mm.	Actual Stress		137.7 N./mm ²
Spec. Min. Thk t _e , or s _a		95.0000 mm.	Spec. Nominal Thk.		95.000 mm.
Surface Area		67669.9 cm ²	SG of Contents		1.000
Element Volume		2428281.5 cm ³	Weight of Contents		2426.8 Kg
Empty Weight		4718.8 Kg	Operating Weight		7145.6 Kg
			Comp. All. Ope. Stress		147.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (147.158 \cdot 805.5) / (138.0 \cdot 1.0 - 0.6 \cdot 147.158)$$

$$= 91.7744 + 3.0000 = 94.7744 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.158 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 92.0) / (805.5 + 0.6 \cdot 92.0)$$

$$= 147 - 0 = 147 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 95.0) / (802.5 + 0.6 \cdot 95.0)$$

$$= 153 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (147.158 \cdot (805.5 + 0.6 \cdot 92.0)) / (1.0 \cdot 92.0)$$

$$= 137.680 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 5.588 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.



Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

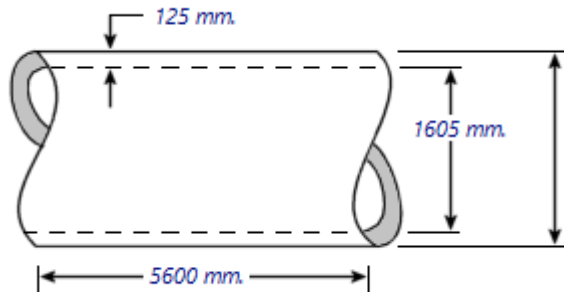
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	191.008 bars	Int. Design Temperature	200 °C	
Material		SA-516 70	Ext. Design Temperature	200 °C	
Length		5600.000 mm.	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1605.000 mm.	Actual Stress	137.6 N./mm ²	
Spec. Min. Thk t,e, or sa		125.0000 mm.	Spec. Nominal Thk.	125.000 mm.	
Surface Area		326348.7 cm ²	SG of Contents	0.050	
Element Volume		7873678.0 cm ³	Weight of Contents	395.0 Kg	
Empty Weight		29486.5 Kg	Operating Weight	29881.6 Kg	
			Comp. All. Ope. Stress	141.2 N./mm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (191.008 \cdot 805.5) / (138.0 \cdot 1.0 - 0.6 \cdot 191.008)$$

$$= 121.5985 + 3.0000 = 124.5985 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.008 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 122.0) / (805.5 + 0.6 \cdot 122.0)$$

$$= 192 - 0.01 = 192 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 125.0) / (802.5 + 0.6 \cdot 125.0)$$

$$= 197 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (191.008 \cdot (805.5 + 0.6 \cdot 122.0)) / (1.0 \cdot 122.0)$$

$$= 137.581 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 7.225 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.



Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

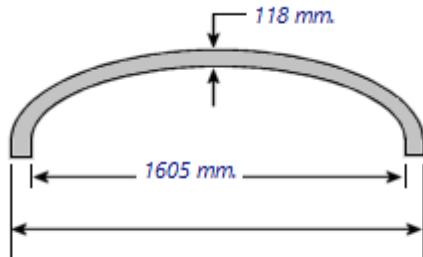
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Dish End

Elliptical Head Design Information:					
Design Pressure	P	191.008 bars	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		50.800 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1605.000 mm.	Actual Stress		135.0 N./mm ²
Spec. Min. Thk t _e , or s _a		118.0000 mm.	Spec. Nominal Thk.		130.000 mm.
Surface Area		40679.9 cm ²	SG of Contents		0.050
Element Volume		644101.6 cm ³	Weight of Contents		32.3 Kg _f
Empty Weight		3909.0 Kg _f	Operating Weight		3941.4 Kg _f
			Comp. All. Ope. Stress		141.2 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 20.062 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.031 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (191.008 \cdot 1611.0 \cdot 0.995) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 191.008)$$

$$= 112.5057 + 3.0000 = 115.5057 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.008 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 115.0) / (0.995 \cdot 1611.0 + 0.2 \cdot 115.0)$$

$$= 195 - 0.01 = 195 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 118.0) / (1.0 \cdot 1605.0 + 0.2 \cdot 118.0)$$

$$= 200 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (191.008 \cdot (0.995 \cdot 1611.0 + 0.2 \cdot 115.0)) / (2 \cdot 1.0 \cdot 115.0)$$

$$= 135.046 \text{ N./mm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (191.008 \cdot 805.5) / (138.0 \cdot 1.0 - 0.6 \cdot 191.008) + 3.0$
 $= 124.598 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.008 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138.0 \cdot 1.0 \cdot 127.0) / (805.5 + 0.6 \cdot 127.0)$
 $= 199 - 0.01 = 199 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (1611.0 / (2 \cdot 404.25))^2) / 6$
 $= 0.995062$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 28.859 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Warning:
 The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
 This component must be impact tested.

MDMT Calculations in the Head Straight Flange:

Warning:
 The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
 This component must be impact tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	191.578 bars
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	130.000 mm.	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 20.062 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.031 mm.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	249.052 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	255.203 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	210.736 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	191.578 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	191.540 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	191.100 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	197.942 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	162.072 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	210.210 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	210.210 bars
App 27-4	= M.A.W.P.	147.339 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	179.4	234.0	0.766	191.70
Main Shell	179.5	234.0	0.767	249.21
Dish End	176.2	234.0	0.753	249.21

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
T2-Tube outlet	Nozzle	138.00	138.00	1.000
T1-Tube Inlet	Nozzle	138.00	138.00	1.000
S1-Shell Inlet	Nozzle	138.00	138.00	1.000
S2-Shell outlet	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Blind Flg	138.00	138.00	1.000
Ch Flg	165.00	153.14	1.077
LHS Ch Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
T2-Tube outlet	96.25	225.00	0.428
T1-Tube Inlet	96.25	225.00	0.428
S1-Shell Inlet	145.38	225.00	0.646
S2-Shell outlet	145.38	225.00	0.646

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	249.052 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	255.203 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	210.736 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	191.578 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	191.540 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	191.100 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	197.942 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	162.072 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	210.210 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	210.210 bars
App 27-4	= M.A.W.P.	147.339 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	185.3	234.0	0.792	198.10
Main Shell	183.9	234.0	0.786	255.36
Dish End	180.5	234.0	0.772	255.36

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
T2-Tube outlet	Nozzle	138.00	138.00	1.000
T1-Tube Inlet	Nozzle	138.00	138.00	1.000
S1-Shell Inlet	Nozzle	138.00	138.00	1.000
S2-Shell outlet	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Blind Flg	138.00	138.00	1.000
Ch Flg	165.00	153.14	1.077
LHS Ch Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
T2-Tube outlet	99.46	225.00	0.442
T1-Tube Inlet	99.46	225.00	0.442
S1-Shell Inlet	148.96	225.00	0.662
S2-Shell outlet	148.96	225.00	0.662

Elements Suitable for Test Pressure.

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MDMT Calculations for Stiffening Rings:

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		255.20 bars
Shell Thickness	ts	125.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1605.000 mm.
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		197.94 bars
Channel Thickness	tc	95.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1605.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1126
Tube Wall Thickness	et	3.4040 mm.
Tube Outside Diameter	D	25.4000 mm.
Total Straight Tube Length	Lt	5150.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		4860.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	40.6400 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.4000 mm.
Groove Weld Leg	ag	3.4000 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	764.093 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	15688.0 cm ²
Perimeter of Tube Layout	Cp	4749.9 mm.

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a



Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-266 2

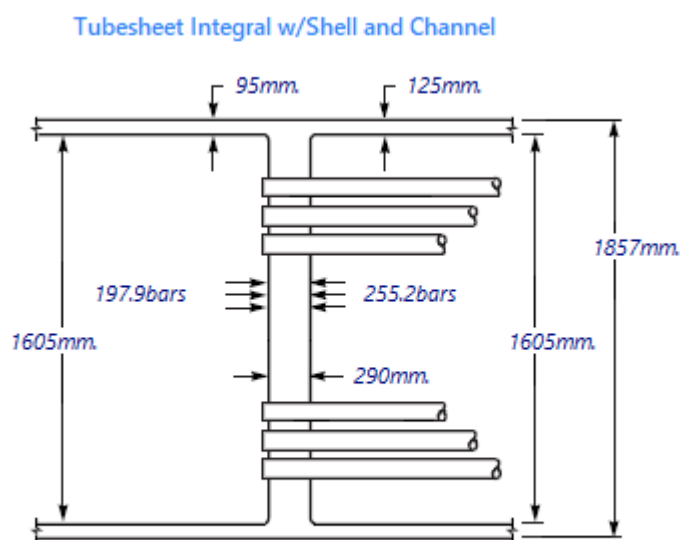
See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 290.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm.
 Tubesheet Outside Diameter A 1857.000 mm.

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Tubesheet Integral with Both
 Tubesheet Extended as Flange No



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Channel - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 21.1 °C 0.20270E+09 KPa.
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (197.94 \cdot (25.4/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 197.94)$
 $= 2.5061 + 0.0000 = 2.5061 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
3.4040	25.40	5624.09	7.46	50.0000	0.0197562	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 198.4858 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
4.0981	25.40	5624.09	6.20	50.0000	0.0286344	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 255.2354 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	4.0981 mm.
Allowable Internal Pressure at Corroded thickness	277.38 bars
Required Internal Design Pressure	197.94 bars
Allowable External Pressure at Corroded thickness	198.49 bars
Required External Design Pressure	255.20 bars
Required Thickness due to Shell Side pressure	4.0981 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot r_o + dt$
 $= 2 \cdot 764.093 + 25.4$
 $= 1553.586 \text{ mm.}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (40.64 - 25.4) / 40.64$
 $= 0.375$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= D_s / D_o$ (Configurations a, b, c)
 $= 1605.0 / 1553.5861$
 $= 1.0331$

Ratio [pc]:

$= D_c / D_o$ (Configurations a, e, f)
 $= 1605.0 / 1553.5861$
 $= 1.0331$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned}
 &= l_{tx} / h \\
 &= 0/290.0 \\
 &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}
 \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned}
 &= \max(dt - 2t(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2t) \\
 &= \max(25.4 - 2 \cdot 3.404 (0.20270E+09/.20270E+09) \cdot \\
 &\quad (92/138) \cdot (0); 25.4 - 2 \cdot 3.404) \\
 &= 25.4000 \text{ mm.}
 \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned}
 &= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\
 &= 40.64 / \sqrt{ (1 - 4 \cdot \min(0 ; 4 \cdot 1553.586 \cdot 40.64) / (\pi \cdot 1553.586^2)) } \\
 &= 40.6400 \text{ mm.}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (40.64 - 25.4) / 40.64 \\
 &= 0.3750
 \end{aligned}$$

$$\begin{aligned}
 h/p &= 7.135827 ; \mu^* = 0.375000 ; E^* = 77090800 \text{ KPa.} \\
 E^*/E &= 0.380317 ; \nu^* = 0.317427
 \end{aligned}$$

Note: As h/p (7.136) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1605.0 + 125.0) 125.0) } \\
 &= 0.0039 \text{ 1/mm.}
 \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2)) \\
 &= 0.004 \cdot 2013223 \cdot 125.0^3 / (6(1 - 0.3^2)) \\
 &= 2815184640.0000 \text{ bars} \cdot \text{mm.}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned}
 &= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\
 &= 6(1605.0 + 125.0) \cdot 0.28152E+1 / 290.0^3 \cdot (1 + 290.0 \cdot 0.004 + (290.0 \cdot 0.004)^2 / 2) \\
 &= 3326311.8 \text{ bars}
 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned}
 &= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s))) \\
 &= 1605.0^2 / (4 \cdot 0.20132E+09 \cdot 125.0) \cdot (1 - 1605.0 \cdot 0.3 / (2(1605.0 + 125.0))) \\
 &= 0.22028E-01 \text{ mm./N./mm}^2
 \end{aligned}$$

Parameter [ω_s]:

$$\begin{aligned}
 &= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\
 &= 1.0331 \cdot 0.28152E+1 \cdot 0.0039 \cdot 0.022028 (1 + 290.0 \cdot 0.0039) \\
 &= 53438.6211 \text{ mm.}^2
 \end{aligned}$$

Channel coefficient [β_c]:

$$\begin{aligned}
 &= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1605.0 + 95.0) 95.0) } \\
 &= 0.0045 \text{ 1/mm.}
 \end{aligned}$$

Channel coefficient [k_c]:

$$\begin{aligned}
 &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2)) \\
 &= 0.005 \cdot 2013223 \cdot 95.0^3 / (6(1 - 0.3^2)) \\
 &= 1430011392.0000 \text{ bars} \cdot \text{mm.}^2
 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$\begin{aligned}
 &= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + \\
 &\quad h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\
 &= 6(1605.0 + 95.0 / \cos(0)) \cdot 0.14300E+1 / 290.0^3 \cdot (1.0 + \\
 &\quad 290.0 \cdot 0.005 \cdot (1.0 + 1.0) / 2 + (290.0 \cdot 0.005)^2 / 2 \cdot 1.0)
 \end{aligned}$$

= 1897174.4 bars

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1605.0^2 / (4 \cdot 0.20132E+09 \cdot 95.0) \cdot (1 - 1605.0 \cdot 0.3 / (2(1605.0 + 95.0)))$$

$$= 0.28902E-01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ωc]:

$$= \rho_c \cdot k_c [\delta c (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta + h \cdot \beta C / 2)]$$

$$= 1.033 \cdot 0.14300E+1 [0.028902 (1.0 + 0.005 \cdot 1.0 \cdot 290.0) - 0 (1.0/290.0 + 1.0 \cdot 0.005/2)]$$

$$= 44653.3516 \text{ mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1857.0 / 1553.5861$$

$$= 1.1953$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.32) / 777090800 \cdot (3326311 + 1897174 + 0.20270E+09 \cdot \ln(1.2))$$

$$= 4.9451$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 1553.586^2 / 16 \cdot [(1.033 - 1) \cdot (1.033^2 + 1) \cdot 255.203 - (1.033 - 1) \cdot (1.033^2 + 1) \cdot 197.942]$$

$$= 590958.3750 \text{ bars} \cdot \text{mm.}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + \omega c \cdot P_t - \omega_s \cdot P_s$$

$$= 590958 + 44653.352 \cdot 197.942 - 53438.621 \cdot 255.203$$

$$= -4207978.5000 \text{ bars} \cdot \text{mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F (P_s - P_t) / (1 + F)$$

$$= (-4207978.5) - 1553.586^2 / 32 \cdot 4.945 (255.2 - 197.94) / (1 + 4.95)$$

$$= -4300314.5000 \text{ bars} \cdot \text{mm.}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= -4300314.5 + 1553.586^2 / 64 \cdot (3 + 0.317) \cdot (255.2 - 197.94)$$

$$= 2863647.0000 \text{ bars} \cdot \text{mm.}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|-4300314.5|; |2863647|)$$

$$= 4300314.5000 \text{ bars} \cdot \text{mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 4300314 / (0.375 (290.0 - 0)^2)$$

$$= 81.8181 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 250.0$$

$$= 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness $[\sigma]$:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 4281052 / (0.375 (120.3594 - 0)^2)$$

$$= 472.8637 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress $[H_{reqB}]$:

$$= h + C_{ats} + C_{atc}$$

$$= 120.3594 + 0 + 0$$

$$= 120.3594 \text{ mm.}$$

Required tubesheet thickness for given loadings (includes CA) $[H_{req}]$:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(120.3594 ; 26.5455)$$

$$= 120.3594 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential $[|P_s - P_t|]$:

$$= |P_s - P_t|$$

$$= |255.2 - 197.94|$$

$$= 57.261 \text{ bars}$$

Shear Stress check $[\tau_{limit}]$:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta_T \cdot S_y$$

$$= 4 \cdot 0.375 \cdot 290.0 / 1553.59 \cdot 0.8 \cdot 0.95 \cdot 249.995$$

$$= 53.199 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}) . Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear $[H_{reqS}]$:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta_T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.375 \cdot 189.997) \cdot (4 \cdot 15688.0 / 4749.92) \cdot |255.2 - 197.94| + 0$$

$$= 26.5455 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 125.000mm.
 Min. Shell len. adjacent to the tubesheet is: 806.241 mm.

Note: For a given Channel thickness of: 95.000 mm.
 Min. Channel len. adjacent to the tubesheet is: 702.865mm.

Tube Metal Area $[A_t]$:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 3.404) 3.404$$

$$= 2.352 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength $[F_t]$:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.404 (25.4 - 3.404) 170.997$$

$$= 4101.500 \text{ Kg}$$

Groove Weld Strength $[F_g]$:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3.4 (25.4 + 0.67 \cdot 3.4) 171.0$$

$$= 4101.5005 \text{ Kg}$$

Weld Strength Factor $[f_w]$:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 170.997 / 170.997$$

$$= 1.000$$

Design Strength Ratio $[f_d]$:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 3.4 (25.4 + 0.67 \cdot 3.4) 171.0$$

$$= 2835.2144 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 3.404 \cdot (25.4 - 3.404) \cdot 1.0 \cdot 1.0)} - 0.75 \cdot 25.4 \right)$$

$$= 3.996 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 2.0424 mm.

Minimum Required Groove Weld Leg [agr]: 2.0424 mm.

Tube-Tubesheet Jt allowable, 4101.5 is $\geq$ tube strength 4101.5 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	81.8 $\leq$	475.0 N./mm ²	Ok
Tubesheet shear stress	17.4 $\leq$	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	120.359	290.000	Ok
Tube-Tubesheet Fillet Weld Leg :	2.042	3.400	Ok
Tube-Tubesheet Groove Weld Leg :	2.042	3.400	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses		Case Type	Pass/Fail
		Bend	Allwd	Shear	Allwd	
D1uc	217.281 ...	258	475	60	190	Ps+Pt D1 Ok
D2uc	246.726 ...	340	475	78	190	Ps+Pt D2 Ok
D3uc	120.359 ...	82	475	17	190	Ps+Pt D3 Ok
Max:	246.7259 ... mm.	0.715	0.408	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	370	...	370	Ok
D2uc	...	...	370	...	370	Ok



D3uc	...	...	...	370	...	370	Ok
Max	...	...	...	...	...	...	...

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	246.726	290.000	Ok
<i>Tube thickness :</i>	<i>4.098</i>	<i>3.404</i>	<i>Fail</i>
Tube-tubesheet fillet weld leg :	2.042	3.400	Ok
Tube-tubesheet groove weld leg :	2.042	3.400	Ok

Given Shell Thickness: 125.0000 mm.
 Given Channel Thickness: 95.0000 mm.

Min Shell length of thk, (125.000) adj. to tubesheet: 806.241 mm.

Min Channel length of thk, (95.000) adj. to tubesheet: 702.865 mm.

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	...	602	No Calc	No Calc
20	30	No Calc	...	582	No Calc	No Calc
30	Tsht	1199.97	1795	92	0.026362	122.731
Tsht	40	No Calc	...	92	No Calc	No Calc
40	50	5784.58	1855	122	0.0066159	110.339
50	60	No Calc	1841	115	0.0086758	113.092

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	605	362.534	...	No Calc
20	30	585	545.668	...	No Calc
30	Tsht	95	No Calc	...	83.867
Tsht	40	95	No Calc	...	No Calc
40	50	125	No Calc	...	96.7513
50	60	118	No Calc	...	78.4891

Minimum

78

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1199.97	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	5784.58	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	42882.5	...	42882.5	...	...
20	30	33392.3	1603635	33299.8	1606515	...
30	40	4718.82	2428281	4577.87	2446470	...
40	50	29486.5	7873678	28828.8	7958560	...
50	60	3909.04	644102	3818.83	650965	...
Total		114389	12549697.00	113407	12662512.00	0



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	...	302.5	...	...	Liquid: 10
20	Liqd	1602.66	387.5	...	...	Liquid: 20
30	Liqd	2426.8	600	...	...	Liquid: 30
30	Nozl	952.153	600	965.212	...	T2-Tube outlet
30	Nozl	1014.37	600	-965.212	...	T1-Tube Inlet
40	Sadl	704.577	755	978.75	...	Left Saddle
40	Sadl	704.577	4823.92	978.75	...	New Saddle
40	Liqd	395.02	2800	...	...	Liquid: 40
40	Insl	321.748	2800	...	...	Hot Insulation
40	Nozl	2586.42	1500	991.425	...	S1-Shell Inlet
40	Nozl	1564.48	3784.72	-991.425	...	S2-Shell outlet
40	Wght	10000	2000	...	...	Additional weig
50	Liqd	32.3142	184.55	...	...	Liquid: 50
50	Insl	45.0347	226.025	...	...	Hot Insulation
30	FTsh	4804.89	1345	...	...	
30	Tube	12399.3	3775	...	...	Tubes
40	Bafl	199.188	1010	...	...	Baffles
40	Bafl	199.188	1450	...	...	Baffles
40	Bafl	199.188	1890	...	...	Baffles
40	Bafl	199.188	2330	...	...	Baffles
40	Bafl	199.188	2770	...	...	Baffles
40	Bafl	199.188	3210	...	...	Baffles
40	Bafl	200.685	3650	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	1409.2
Liquid	4456.8
Insulation	366.8
Nozzles	6117.4
Weights	10000.0
Exchanger Components	18600.0
Liquid in Tubes	1940.4

Sum of the Detail Weights 42890.5 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	114389.2	114389.2	114389.2	114389.2	114389.2	114389.2
Saddles	1409.2	1409.2	1409.2	1409.2	1409.2	1409.2
Nozzles	6117.4	6117.4	6117.4	6117.4	6117.4	6117.4



Wld Weights	10000.0	10000.0	10000.0	10000.0	10000.0	10000.0
Exchanger	18600.0	18600.0	18600.0	18600.0	18600.0	18600.0
Insulation	...	...	366.8	366.8	366.8	366.8
Ope. Liquid	...	...	...	...	...	4456.8
Tube Ope Lqd	...	...	...	...	...	1940.4
Test Liquid	...	12542.0	...	...	...	...
Tube Tst Lqd	...	1940.4	...	...	...	...
Totals	150515.8	164998.1	150882.5	150882.5	150882.5	157279.7

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	150515.8 kg.
Shop Test Wt.	- Fabricated weight + water (full)	164998.1 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	150882.5 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	150882.5 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	150882.5 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	157279.7 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	156298.2 kg.
Field Test Wt.	- Empty weight + test medium (full)	164496.2 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	1941547.9 cm ³
Weight of Ope Liq in tubes :	1940.4 kg.
Weight of Water in tubes :	1940.4 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	155543	...
20	30	144965	39733
30	40	67669.9	60507.1
40	50	326349	282366
50	60	40679.9	30485.2

Total	735206.9	413091.6 cm ²
Total	791.4	444.6 ft ²

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
T2-Tube outlet	238.12	255.10	85	1500	GR 1.1	...	...	...	...
T1-Tube Inlet	238.12	255.10	85	1500	GR 1.1	...	...	...	...
S1-Shell Inlet	219.10	255.10	200	1500	GR 1.1	...	...	...	...
S2-Shell outle	219.10	255.10	200	1500	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 219.101 bars
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 255.100 bars

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 238.123 bars
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 255.100 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	0 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.194 \cdot 0.983) / (1 + 1.7 \cdot 3.4 \cdot 0.194)))$
 $= \min(0.85, 0.917)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.763
Structure Height to Diameter ratio	4.762

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$: $Zg = 589788$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1300.0 + 0.0 = 1300.0$ mm.
 $= 4.265$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (4.265 ft.) < 15 ft.
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15 / 1935.0)^{2/11.5}$
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:
 $K1 = 0$, $K2 = 0$, $K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.763 \cdot 26.491$$

$$= 569.3 \text{ lbs. } [258.2 \text{ Kg}]$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Blind Flg	1300.0	0.000	0.000	0.000	1.035	1.000	161.849
Ch Flg	1300.0	0.000	0.000	0.000	1.035	1.000	161.849
LHS Ch Shell	1300.0	0.000	0.000	0.000	1.035	1.000	161.849
Main Shell	1300.0	0.000	0.000	0.000	1.035	1.000	161.849
Dish End	1300.0	0.000	0.000	0.000	1.035	1.000	161.849

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Additional we	0.00	1300.00	161.85	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1300	4068	24611.4	161.849	284.063
20	30	1300	1926	14926.5	161.849	172.281
30	40	1300	2154	25848	161.849	298.336
40	50	1300	2346	131376	161.849	1516.34
50	60	1300	2329.2	9861.21	161.849	113.818

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22
Importance Factor		1.500
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(9.826^{10.75})$
 = 0.111 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.111$, 0.03) per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.217 \cdot 154889 \cdot 1.5$
 = 15148.150 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.217 \cdot 154889$
 = 6732.5 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1695	22127	3944.1
20	30	802.5	22127	1867.34
30	40	802.5	22127	1867.34
40	Sad1	802.5	22127	1867.34
Sad1	50	802.5	22127	1867.34
40	50	802.5	22127	1867.34



50	60	802.5	22127	1867.34
----	----	-------	-------	---------

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5662.635 mm.
Center of Gravity of the Liquid	2003.522 mm.
Center of Gravity of the Insulation	6044.719 mm.
Center of Gravity of the Nozzles	4189.177 mm.
Center of Gravity of the Added Weights (Operating)	4873.174 mm.
Center of Gravity of the Added Weights (Empty)	4873.174 mm.
Center of Gravity of the Tubesheet(s)	2728.175 mm.
Center of Gravity of the Tubes	5158.175 mm.
Center of Gravity of the Baffles	5204.590 mm.
Center of Gravity of Bare Shell New and Cold	2246.003 mm.
Center of Gravity of Bare Shell Corroded	2222.357 mm.
Vessel CG in the Operating Condition	2814.214 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	2822.094 mm.
Vessel CG in the Test Condition	2977.955 mm.

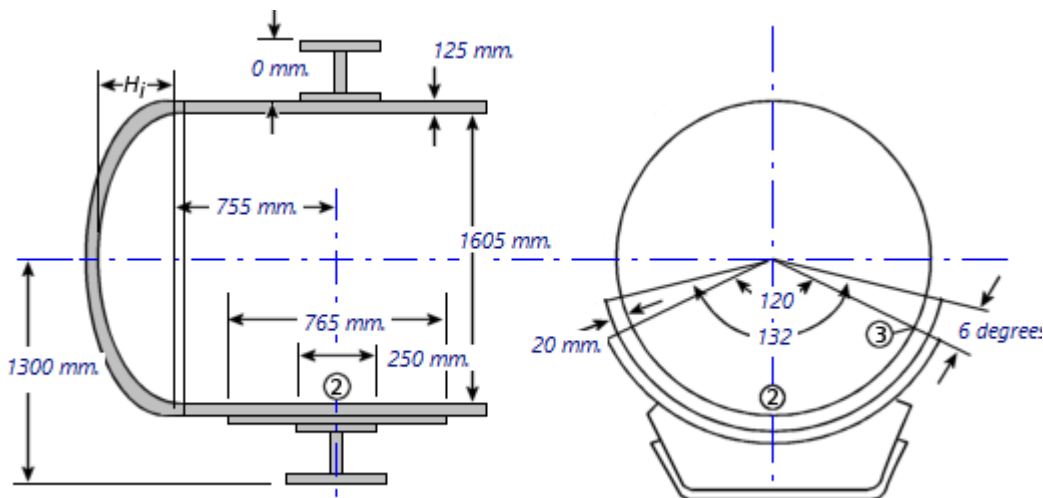
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	866.50	mm.
Shell Thickness used in this Case	t	125.000	mm.
Stiffened Vessel Length per 4.15.6	L	5650.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	755.00	mm.
Saddle Width	b1 or b	250.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	765.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		153.14	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	209283.64	Kgf
Pressure used in Analysis	P 191.004	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	62.67	138.00
Long. Stress at Bottom of Midspan	73.00	138.00
Long. Stress at Top of Saddles	70.14	138.00
Long. Stress at Bottom of Saddles	66.56	138.00
Tangential Shear in Shell	16.66	110.40
Circ. Stress at Horn of Saddle	12.76	172.50
Circ. Compressive Stress in Shell	1.45	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3.0(2384.8/2 + 0) \cdot 1300.0/1550.0$$

$$= 3000.3 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(692.86, 0) \cdot 1300.0/4068.9207$$

$$= 221.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(15148.15, 0) \cdot 1300.0/4068.9207$$

$$= 4839.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3.0(15148/2 + 0) \cdot 1300.0/1550.0$$

$$= 19057.4 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 190226 + \max(221, 3000, 4840, 19057)$$

$$= 209283.6 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(0.0 + \pi/4(1.855 \cdot 1.2)^2)$$

$$= 6794.525 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	209988.22	Kgf
Transverse Shear Load	7574.08	Kgf
Longitudinal Shear Load	15148.15	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1550.000 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0427	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0351			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -209284 \cdot 755.0 [1 - (1 - 755.0/5650.8 + (866.5^2 - 0.0^2)/(2 \cdot 755.0 \cdot 5650.8))/(1 + (4 \cdot 0.0)/(3 \cdot 5650.8))]$$

$$= -7207.9 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 209284 \cdot 5651/4((1 + 2(866.5^2 - 0^2)/(5651^2))/(1 + (4 \cdot 0)/(3 \cdot 5651)) - 4 \cdot 755/5651)$$

$$= 151552.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 191.004 \cdot 866.5/(2 \cdot 122.0) - 151552.6/(\pi \cdot 866.5^2 \cdot 122.0)$$

$$= 62.67 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 191.004 \cdot 866.5/(2 \cdot 122.0) + 151552.6/(\pi \cdot 866.5^2 \cdot 122.0)$$

$$= 73.00 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 191.004 \cdot 866.5 / (2 \cdot 122.0) - 7207.9 / (0.1066 \cdot \pi \cdot 866.5^2 \cdot 122.0)$$

$$= 70.14 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 191.004 \cdot 866.5 / (2 \cdot 122.0) + 7207.9 / (0.1923 \cdot \pi \cdot 866.5^2 \cdot 122.0)$$

$$= 66.56 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm / 3))$$

$$= 209284(5650.8 - 2 \cdot 755.0) / (5650.8 + (4 \cdot 0.0 / 3))$$

$$= 153359.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 153359.11 / (866.5 \cdot 122.0)$$

$$= 16.66 \text{ N./mm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(866.5 \cdot 122.0)}$$

$$= 253.606 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(866.5 \cdot 122.0)}, 2 \cdot 755.0)$$

$$= 757.21 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1)$$

$$= \min(137.997/137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma6, r$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 209284 \cdot 0.1 / (757.211(122.0 + 1.0 \cdot 20.0))$$

$$= -1.45 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [$\sigma7, r^*$]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -209284 / (4(122.0 + 1.0 \cdot 20.0)757.211) -$$

$$12 \cdot 0.043 \cdot 209284 \cdot 866.5 / (5650.8(122.0 + 1.0 \cdot 20.0)^2)$$

$$= -12.76 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 4068.9 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000013 \cdot 4068.921(200.0 - 21.1)$$

$$= 9.280 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1550.0000	mm.
Baseplate Thickness	Bpthk	52.0000	mm.
Baseplate Width	Bpwid	275.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm.
Web Thickness	Webtk	20.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	240.3	N./mm ²
Height of Web at Center	Hw,c	500.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(805.5 \cdot 122.0) + 765.0})^3) / 12$$

$$= 18976.078 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 18976.078 + 1529.919 (105.026 - 61.0)^2$$

$$= 48630.113 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1254.0	122.0	61.0	1529.9	9332505.0	0.486E+05
Wearplate	765.0	20.0	132.0	153.0	2019599.8	0.116E+04
Web	20.0	300.5	292.2	60.1	1756421.9	0.256E+05
BasePlate	275.0	52.0	468.5	143.0	6699548.5	0.189E+06
Totals	...	...	...	1886.0	19808076.0	0.265E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 19808076 / 1886.019$$

$$= 105.026 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 209283.641$$

$$= 42593.7578 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 11.7301 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 144.1974 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1300.0 - 805.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 633.857 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 26998.8789 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 10.5081 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 160.2194 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(209284 + 705) \cdot 275.0) / (4 \cdot 1550.0 \cdot 160.219)}$$

$$= 41.356 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (802.5 + 125.0 + 20.0)$$

$$= 1641.118 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 1641.118 / (5 - 1)$$

$$= 410.280 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 410.2795 \cdot 275.0 / 2$$

$$= 564.134 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 209988.234 / (1550.0 \cdot 275.0)$$

$$= 49.264 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 564.1 \cdot 49.26$$

$$= 27791.568 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 46.0 + 41.028$$

$$= 87.028 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 27791.6 / 87.0279$$

$$= 31.317 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	250.0	...	50.0	...	0.260E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 250.0 - 20.0$$

$$= 230.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 87.028$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 250.0 / 2$$

$$= 125.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (2604.2 / 87.028) }$$

$$= 54.702 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk) - bpthk$$

$$= 1300.0 - \cos(120.0 / 2)(802.5 + 125.0 + 20.0) - 52.0$$

$$= 774.250 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk)$$

$$= 2 \cdot \sin(120.0 / 2)(802.5 + 125.0 + 20.0)$$

$$= 1641.118 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= Baseplate \text{ length} - clearance$$

$$= 1550.0 - 50.8$$

$$= 1499.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (1641.118 - 1499.2) / 2$$

$$= 70.959 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{774.25^2 + 70.959^2}$$

$$= 777.495 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 240.3}$$

$$= 128.153$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 777.495 / 54.702$$

$$= 14.213$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 15148.2 / (2 \cdot 1550.0) \cdot 410.28 \cdot 777.49 / 2$$

$$= 779.388 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($14.2132 < 128.1526$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.21)^2 / (2 \cdot 128.15^2)) 240 / (5/3 + 3 \cdot (14.21) / (8 \cdot 128.15) - (14.21)^3 / (8 \cdot 128.15^3))$$

$$= 139.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 31.32 / 139.84 + (779.39 \cdot 125.0 / 26041666) / 160.22$$

$$= 0.453$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	230.0	0.0	46.0	0.0	0.260E+04
Web	410.3	20.0	0.0	82.1	0.0	27.4
Totals	...	...	...	128.1	...	0.263E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 128.056$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0 / 2$$

$$= 125.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1300.0 - \cos(947.5/2) (52.0 + 0.0 + 0.0) - 0.0$$

$$= 300.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{2630.2 / 128.056}$$

$$= 45.320 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 300.5 / 45.32$$

$$= 6.631$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 15148.15 / (2 \cdot 1550.0)$$

$$= 4.887 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.887 \cdot 410.28 \cdot 300.5$$

$$= 602.464 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 602.464 / 210414.812$$

$$= 28.079 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.6306 < 128.1526$) per AISC E2-1 [Sca]:

$$= (1 - (K/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (K/r) / (8 \cdot C_c) - (K/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.63)^2 / (2 \cdot 128.15^2)) 240 /$$

$$(5/3 + 3 \cdot (6.63) / (8 \cdot 128.15) - (6.63)^3 / (8 \cdot 128.15^3))$$

$$= 142.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 42.59 / 142.35 + (602.46 \cdot 125.0 / 2630.183) / 160.22$$

$$= 0.474$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	149.95 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	9.7967 cm ²
Saddle Load QO (Weight)	QO	190930.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	4839.8 Kg
Maximum Transverse Force	Ft	7574.1 Kg
Maximum Longitudinal Force	Fl	15148.2 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 2770.6 \text{ Kg}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 15148.2 / (9.797 \cdot 4)$$

$$= 37.9 \text{ N./mm}^2. \text{ Must be less than } 99.9 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 7574.1 / (9.797 \cdot 4)$$

$$= 19.0 \text{ N./mm}^2. \text{ Must be less than } 99.9 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 15148.15 / (99.93 \cdot 4.0)$$

$$= 3.7164 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1300.0 \cdot 7574.08 + 0.0$$

= 9846.49 Kg-m.

Eccentricity (e):

= Rmom / QO

= 9846.49/190930.86

= 51.57 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 3.7164, 0.0]

= 3.7164 cm²

Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	866.50	mm.
Shell Thickness used in this Case	t	125.000	mm.
Stiffened Vessel Length per 4.15.6	L	5650.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	755.00	mm.
Saddle Width	b1 or b	250.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	765.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	461.75	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm.
Coefficient of Friction	mu	0.08	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		52985.44	Kgf
Pressure used in Analysis	P	191.004	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	66.82	138.00
Long. Stress at Bottom of Midspan	68.85	138.00
Long. Stress at Top of Saddles	69.90	138.00
Long. Stress at Bottom of Saddles	66.69	138.00
Tangential Shear in Shell	3.80	110.40
Circ. Stress at Horn of Saddle	3.23	172.50
Circ. Compressive Stress in Shell	0.37	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3.0(2384.8/2 + 0) • 1300.0/1550.0

= 3000.3 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(692.86, 0) \cdot 1300.0/4068.9207$
 $= 221.4 \text{ Kg}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(15148.15, 0) \cdot 1300.0/4068.9207$
 $= 4839.8 \text{ Kg}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(15148/2 + 0) \cdot 1300.0/1550.0$
 $= 19057.4 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 33928 + \max(221, 3000, 4840, 19057)$
 $= 52985.4 \text{ Kg}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4)(OD \cdot \text{WindDiaMult})^2)$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(1.855 \cdot 1.2)^2)$
 $= 6794.525 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	53690.01	Kgf
Transverse Shear Load	7574.08	Kgf
Longitudinal Shear Load	15148.15	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 1550.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0427	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0351			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -52985 \cdot 755.0 [1 - (1 - 755.0/5650.8 + (866.5^2 - 461.75^2)/(2 \cdot 755.0 \cdot 5650.8))/(1 + (4 \cdot 461.75)/(3 \cdot 5650.8))]$
 $= -6477.4 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 52985 \cdot 5651/4((1 + 2(867^2 - 462^2)/(5651^2))/(1 + (4 \cdot 462)/(3 \cdot 5651)) - 4 \cdot 755/5651)$
 $= 29767.9 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 191.004 \cdot 866.5/(2 \cdot 122.0) - 29767.9/(\pi \cdot 866.5^2 \cdot 122.0)$
 $= 66.82 \text{ N/mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 191.004 \cdot 866.5/(2 \cdot 122.0) + 29767.9/(\pi \cdot 866.5^2 \cdot 122.0)$
 $= 68.85 \text{ N/mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:
 $= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$



$$= 191.004 \cdot 866.5 / (2 \cdot 122.0) - 6477.4 / (0.1066 \cdot \pi \cdot 866.5^2 \cdot 122.0)$$

$$= 69.90 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^* 4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 191.004 \cdot 866.5 / (2 \cdot 122.0) + 6477.4 / (0.1923 \cdot \pi \cdot 866.5^2 \cdot 122.0)$$

$$= 66.69 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 52985(5650.8 - 2 \cdot 755.0) / (5650.8 + (4 \cdot 461.75 / 3))$$

$$= 35012.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 35012.09 / (866.5 \cdot 122.0)$$

$$= 3.80 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(866.5 \cdot 122.0)}$$

$$= 253.606 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(866.5 \cdot 122.0)}, 2 \cdot 755.0)$$

$$= 757.21 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r / S, 1)$$

$$= \min(137.997 / 137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 52985 \cdot 0.1 / (757.211(122.0 + 1.0 \cdot 20.0))$$

$$= -0.37 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -52985 / (4(122.0 + 1.0 \cdot 20.0)757.211) -$$

$$12 \cdot 0.043 \cdot 52985 \cdot 866.5 / (5650.8(122.0 + 1.0 \cdot 20.0)^2)$$

$$= -3.23 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1550.0000	mm.
Baseplate Thickness	Bpthk	52.0000	mm.
Baseplate Width	Bpwid	275.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm.
Web Thickness	Webtk	20.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	240.3	N./mm ²
Height of Web at Center	Hw,c	500.0	mm.
Friction Coefficient	mu	0.080	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(805.5 \cdot 122.0)} + 765.0)^3) / 12$$

$$= 18976.078 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 18976.078 + 1529.919(105.026 - 61.0)^2$$

$$= 48630.113 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1254.0	122.0	61.0	1529.9	9332505.0	0.486E+05
Wearplate	765.0	20.0	132.0	153.0	2019599.8	0.116E+04
Web	20.0	300.5	292.2	60.1	1756421.9	0.256E+05
BasePlate	275.0	52.0	468.5	143.0	6699548.5	0.189E+06
Totals	...	...	...	1886.0	19808076.0	0.265E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 19808076/1886.019$$

$$= 105.026 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 52985.438$$

$$= 10783.6846 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 2.9698 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 144.1974 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1300.0 - 805.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 633.857 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 6835.4473 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.6604 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 160.2194 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(52985 + 705) \cdot 275.0) / (4 \cdot 1550.0 \cdot 160.219)}$$

$$= 20.911 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (802.5 + 125.0 + 20.0)$$

$$= 1641.118 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 1641.118 / (5 - 1)$$

$$= 410.280 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 410.2795 \cdot 275.0/2$$

$$= 564.134 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 53690.016 / (1550.0 \cdot 275.0)$$

$$= 12.596 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 564.1 \cdot 12.6$$

$$= 7105.778 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 46.0 + 41.028$$

$$= 87.028 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 7105.8/87.0279$$

$$= 8.007 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	250.0	...	50.0	...	0.260E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 250.0 - 20.0$$

$$= 230.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/87.028$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0/2$$

$$= 125.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2604.2/87.028)}$$

$$= 54.702 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1300.0 - \cos(120.0/2) (802.5 + 125.0 + 20.0) - 52.0$$

$$= 774.250 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0/2) (802.5 + 125.0 + 20.0)$$

$$= 1641.118 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1550.0 - 50.8$$

$$= 1499.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (1641.118 - 1499.2) / 2$$

$$= 70.959 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{774.25^2 + 70.959^2}$$

$$= 777.495 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994E+09/240.3)}$$

$$= 128.153$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 777.495/54.702$$

$$= 14.213$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 15148.2 / (2 \cdot 1550.0) \cdot 410.28 \cdot 777.49/2$$

$$= 779.388 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (14.2132 < 128.1526) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (14.21)^2 / (2 \cdot 128.15^2)) 240 /$$

$$(5/3 + 3 \cdot (14.21) / (8 \cdot 128.15) - (14.21^3) / (8 \cdot 128.15^3))$$

$$= 139.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 8.01/139.84 + (779.39 \cdot 125.0/26041666) / 160.22$$

$$= 0.286$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	230.0	0.0	46.0	0.0	0.260E+04
Web	410.3	20.0	0.0	82.1	0.0	27.4
Totals	...	...	...	128.1	...	0.263E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/128.056$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0/2$$

$$= 125.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1300.0 - \cos(947.5/2) (52.0 + 0.0 + 0.0) - 0.0$$

$$= 300.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (2630.2/128.056) }$$

$$= 45.320 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 300.5/45.32$$

$$= 6.631$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 15148.15 / (2 \cdot 1550.0)$$

$$= 4.887 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.887 \cdot 410.28 \cdot 300.5$$

$$= 602.464 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 602.464/210414.812$$

= 28.079 N./mm²

Compressive Allowable, $KL/r < Cc$ (6.6306 < 128.1526) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (6.63)^2 / (2 \cdot 128.15^2)) 240 / (5/3 + 3 \cdot (6.63) / (8 \cdot 128.15) - (6.63^3) / (8 \cdot 128.15^3))$$

$$= 142.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 10.89/142.35 + (602.46 \cdot 125.0/2630.183) / 160.22$$

$$= 0.252$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	149.95 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	9.7967 cm ²
Saddle Load QO (Weight)	QO	34632.7 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	4839.8 Kg
Maximum Transverse Force	Ft	7574.1 Kg
Maximum Longitudinal Force	Fl	15148.2 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 15148.2 / (9.797 \cdot 4)$$

$$= 37.9 \text{ N./mm}^2. \text{ Must be less than } 99.9 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 7574.1 / (9.797 \cdot 4)$$

$$= 19.0 \text{ N./mm}^2. \text{ Must be less than } 99.9 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 15148.15 / (99.93 \cdot 4.0)$$

$$= 3.7164 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1300.0 \cdot 7574.08 + 0.0$$

$$= 9846.49 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 9846.49 / 34632.66$$

$$= 284.31 \text{ mm.} > B_{plen}/6 \text{ --> Uplift in Transverse direction}$$

f = Bplen / 2 - Edgedis

$$= 1550.0/2 - 50.8$$

$$= 724.20 \text{ mm.}$$

Modular Ratio Of Steel/Concrete (n1):

$$= ES / EC$$

$$= 203402.5/21526.32$$

$$= 9.45$$

$$\begin{aligned} K1 &= 3 (e - 0.5 \cdot B_{plen}) \\ &= 3 (284.31 - 0.5 \cdot 1550.0) \\ &= -1472.08 \text{ mm.} \end{aligned}$$

$$\begin{aligned} K2 &= 6 \cdot n1 \cdot At / B_{pwid} \cdot (f + e) \\ &= 6 \cdot 9.45 \cdot 19.59 / 275.0 \cdot (724.2 + 284.31) \\ &= 407373.59 \text{ mm.}^2 \end{aligned}$$

$$\begin{aligned} K3 &= -K2 \cdot (0.5 \cdot B_{plen} + f) \\ &= -407373.59 \cdot (0.5 \cdot 1550.0 + 724.2) \\ &= -610734485.27 \text{ mm.}^3 \end{aligned}$$

Iteratively Solving for the Effective Bearing Length:

$$\begin{aligned} Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 &= 0 \\ Y^3 + -1472.08 \cdot Y^2 + 407373.59 \cdot Y + -0.6E+09 &= 0 \\ Y &= 1476.35 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Num} &= (B_{plen} / 2 - Y / 3 - e) \\ &= (1550.0 / 2 - 1476.35 / 3 - 284.31) \\ &= -1.42 \end{aligned}$$

$$\begin{aligned} \text{Denom} &= (B_{plen} / 2 - Y / 3 + f) \\ &= (1550.0 / 2 - 1476.35 / 3 + 724.2) \\ &= 1007.08 \end{aligned}$$

$$\begin{aligned} \text{Total Bolt Tension Force [Tforce]:} \\ &= -QO \cdot \text{Num} / \text{Denom} \\ &= -34632.66 \cdot -1.42 / 1007.08 \\ &= 48.95 \text{ Kg} \end{aligned}$$

$$\begin{aligned} \text{Bolt Area Required due to Transverse Load [Bltareart]:} \\ &= T_{force} / (Stba \cdot Nbt) \\ &= 48.95 / (149.95 \cdot 2.0) \\ &= 0.0160 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Required Area of a Single Bolt [Bltarear]:} \\ &= \max[Bltarearl, Bltarears, Bltareart] \\ &= \max[0.0, 3.7164, 0.016] \\ &= 3.7164 \text{ cm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned} \text{Bearing Pressure (fc)} \\ &= 2(QO + T_{force}) / (Y \cdot B_{pwid}) \\ &= 2(34632.66 + 48.95) / (1476.35 \cdot 275.0) \\ &= 16.75 \text{ bars} \end{aligned}$$

$$\begin{aligned} \text{Distance from Baseplate Edge to the Web [ADIST]:} \\ &= (B_{plen} - WebLength) / 2 \\ &= (1550.0 - 1499.2) / 2 \\ &= 25.4000 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Overturning Moment due To Bolt Tension [Mt]:} \\ &= T_{force} \cdot Adist \\ &= 48.95 \cdot 25.4 \\ &= 1.24 \text{ Kg-m.} \end{aligned}$$

$$\begin{aligned} \text{Equivalent Bearing Pressure (f1):} \\ &= fc \cdot (Y - Adist) / Y \\ &= 16.75 \cdot (1476.35 - 25.4) / 1476.35 \\ &= 16.47 \text{ bars} \end{aligned}$$

$$\begin{aligned} \text{Overturning Moment due to Bearing Pressure [Mc]:} \\ &= (Adist^2 \cdot B_{pwid} / 6) \cdot (f1 + 2 \cdot fc) \\ &= (25.4^2 \cdot 275.0 / 6) \cdot (16.47 + 2 \cdot 16.75) \\ &= 15.07 \text{ Kg-m.} \end{aligned}$$



Baseplate Required Thickness [Treq]:
= $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
= $\sqrt{(6 \cdot \max(1.24, 15.07) / (275.0 \cdot 207.0))}$
= 3.9467 mm.

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

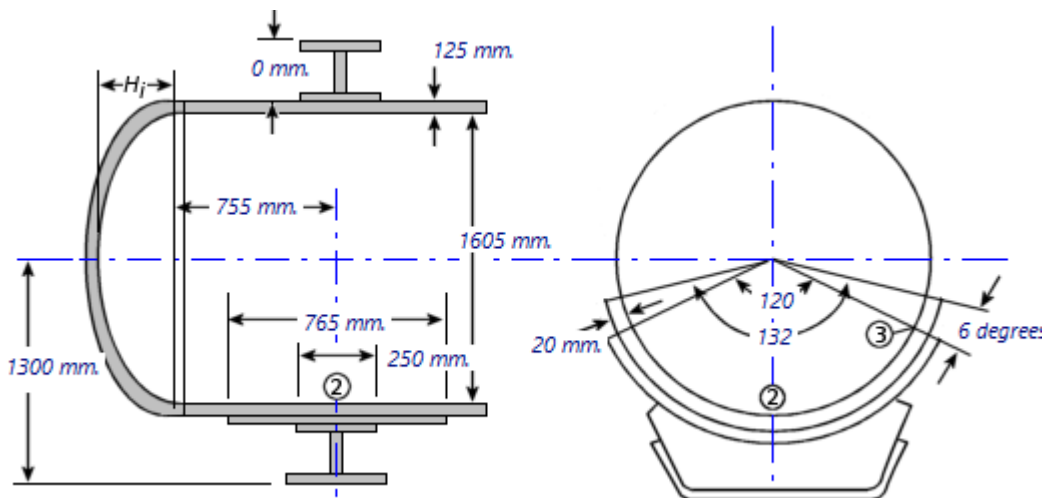
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	866.50	mm.
Shell Thickness used in this Case	t	125.000	mm.
Stiffened Vessel Length per 4.15.6	L	5650.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	755.00	mm.
Saddle Width	b1 or b	250.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	765.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	P	26715.49	Kgf
Pressure used in Analysis		255.288	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	90.00	234.00
Long. Stress at Bottom of Midspan	91.32	234.00
Long. Stress at Top of Saddles	90.96	234.00
Long. Stress at Bottom of Saddles	90.50	234.00
Tangential Shear in Shell	2.13	138.58
Circ. Stress at Horn of Saddle	1.63	350.99
Circ. Compressive Stress in Shell	0.19	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(787.0/2 + 0) \cdot 1300.0/1550.0$$

$$= 990.1 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(228.64, 0) \cdot 1300.0/4068.9207$$

$$= 73.1 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 25725 + \max(73, 990, 0, 0)$$

$$= 26715.5 \text{ Kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(0.0 + \pi/4(1.855 \cdot 1.2)^2)$$

$$= 6794.525 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	27420.06	Kgf
Transverse Shear Load	393.50	Kgf
Longitudinal Shear Load	228.64	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1550.000 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0427	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0351			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 \cdot hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -26715 \cdot 755.0 [1 - (1 - 755.0/5650.8 + (866.5^2 \cdot 0.0^2) / (2 \cdot 755.0 \cdot 5650.8)) / (1 + (4 \cdot 0.0) / (3 \cdot 5650.8))]$$

$$= -920.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 \cdot hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 26715 \cdot 5651 / 4 ((1 + 2(866.5^2 \cdot 0^2)/(5651^2)) / (1 + (4 \cdot 0) / (3 \cdot 5651)) - 4 \cdot 755 / 5651)$$

$$= 19346.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 255.288 \cdot 866.5 / (2 \cdot 122.0) - 19346.0 / (\pi \cdot 866.5^2 \cdot 122.0)$$

$$= 90.00 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 255.288 \cdot 866.5 / (2 \cdot 122.0) + 19346.0 / (\pi \cdot 866.5^2 \cdot 122.0)$$

$$= 91.32 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 255.288 \cdot 866.5 / (2 \cdot 122.0) - 920.1 / (0.1066 \cdot \pi \cdot 866.5^2 \cdot 122.0)$$

$$= 90.96 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 255.288 \cdot 866.5 / (2 \cdot 122.0) + 920.1 / (0.1923 \cdot \pi \cdot 866.5^2 \cdot 122.0)$$

$$= 90.50 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a)/(L+(4 \cdot hm/3))$
 $= 26715(5650.8 - 2 \cdot 755.0)/(5650.8 + (4 \cdot 0.0/3))$
 $= 19576.6 \text{ Kgf}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K_2 \cdot T / (R_m \cdot t)$
 $= 1.1707 \cdot 19576.61 / (866.5 \cdot 122.0)$
 $= 2.13 \text{ N./mm}^2$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(866.5 \cdot 122.0)}$
 $= 253.606 \text{ mm.}$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$
 $= \min(250.0 + 1.56 \cdot \sqrt{(866.5 \cdot 122.0)}, 2 \cdot 755.0)$
 $= 757.21 \text{ mm.}$

Wear Plate/Shell Stress ratio (4.15.26) [η]:
 $= 1.0000$ Materials are the same, test case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:
 $= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$
 $= -0.7603 \cdot 26715 \cdot 0.1 / (757.21(122.0 + 1.0 \cdot 20.0))$
 $= -0.19 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:
 $= -Q/(4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$
 $= -26715/(4(122.0 + 1.0 \cdot 20.0)757.21) -$
 $12 \cdot 0.043 \cdot 26715 \cdot 866.5 / (5650.8(122.0 + 1.0 \cdot 20.0)^2)$
 $= -1.63 \text{ N./mm}^2$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1550.0000	mm.
Baseplate Thickness	Bpthk	52.0000	mm.
Baseplate Width	Bpwid	275.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm.
Web Thickness	Webtk	20.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	500.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(805.5 \cdot 122.0)} + 765.0)122.0^3)/12$
 $= 18976.078 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$
 $= 18976.078 + 1529.919(105.026 - 61.0)^2$
 $= 48630.113 \text{ cm}^4$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1254.0	122.0	61.0	1529.9	9332505.0	0.486E+05
Wearplate	765.0	20.0	132.0	153.0	2019599.8	0.116E+04
Web	20.0	300.5	292.2	60.1	1756421.9	0.256E+05
BasePlate	275.0	52.0	468.5	143.0	6699548.5	0.189E+06
Totals	...	...	...	1886.0	19808076.0	0.265E+06

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 19808076/1886.019 \\
 &= 105.026 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Angle } [\beta]: \\
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120.0/2 \\
 &= 120.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Coefficient [K1]:} \\
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Force [Fh]:} \\
 &= K1 \cdot Q \\
 &= 0.204 \cdot 26715.486 \\
 &= 5437.1807 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 1.4974 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\
 &= 1300.0 - 805.5 \cdot \sin(120.0/2) / 1.0472 \\
 &= 633.857 \text{ mm.}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 3446.4622 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 1.3414 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}) } \\
 &= \sqrt{ 3(26715 + 705)275.0 / (4 \cdot 1550.0 \cdot 173.33) } \\
 &= 14.368 \text{ mm.}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120.0/2) (802.5 + 125.0 + 20.0) \\
 &= 1641.118 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 1641.118 / (5 - 1) \\
 &= 410.280 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot \text{Bpwid} / 2 \\
 &= 410.2795 \cdot 275.0/2 \\
 &= 564.134 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 27420.064 / (1550.0 \cdot 275.0) \\
 &= 6.433 \text{ Kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 564.1 \cdot 6.43 \\
 &= 3628.997 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of the Rib and Web [Ar]:} \\
 &= \text{Rib Area} + \text{Web Area} \\
 &= 46.0 + 41.028 \\
 &= 87.028 \text{ cm}^2
 \end{aligned}$$

$$\text{Compressive Stress [Sc]:}$$

$$\begin{aligned}
 &= P/Ar \\
 &= 3629.0/87.0279 \\
 &= 4.089 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	250.0	...	50.0	...	0.260E+04

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 250.0 - 20.0 \\
 &= 230.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0.0/87.028 \\
 &= 0.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 250.0/2 \\
 &= 125.000 \text{ mm.}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(2604.2/87.028)} \\
 &= 54.702 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1300.0 - \cos(120.0/2)(802.5 + 125.0 + 20.0) - 52.0 \\
 &= 774.250 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(120.0/2)(802.5 + 125.0 + 20.0) \\
 &= 1641.118 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 1550.0 - 50.8 \\
 &= 1499.200 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (Lw - Bd) / 2 \\
 &= (1641.118 - 1499.2) / 2 \\
 &= 70.959 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \sqrt{(Lv^2 + Dd^2)} \\
 &= \sqrt{(774.25^2 + 70.959^2)} \\
 &= 777.495 \text{ mm.}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)} \\
 &= 123.211
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 777.495/54.702 \\
 &= 14.213
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\
 &= 228.6 / (2 \cdot 1550.0) \cdot 410.28 \cdot 777.49/2
 \end{aligned}$$

= 11.764 Kg-m.

Compressive Allowable, $KL/r < Cc$ (14.2132 < 123.2106) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.21)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (14.21) / (8 \cdot 123.21) - (14.21^3) / (8 \cdot 123.21^3))$$

$$= 151.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.09/151.06 + (11.76 \cdot 125.0/26041666) / 173.33$$

$$= 0.030$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	230.0	0.0	46.0	0.0	0.260E+04
Web	410.3	20.0	0.0	82.1	0.0	27.4
Totals	...	...	...	128.1	...	0.263E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/128.056$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0/2$$

$$= 125.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1300.0 - \cos(947.5/2) (52.0 + 0.0 + 0.0) - 0.0$$

$$= 300.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (2630.2/128.056) }$$

$$= 45.320 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 300.5/45.32$$

$$= 6.631$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 228.644 / (2 \cdot 1550.0)$$

$$= 0.074 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.074 \cdot 410.28 \cdot 300.5$$

$$= 9.093 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 9.093/210414.812$$

$$= 0.424 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (6.6306 < 123.2106) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (6.63)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (6.63) / (8 \cdot 123.21) - (6.63^3) / (8 \cdot 123.21^3))$$

$$= 153.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1 / I) / Sba$$

$$= 5.56/153.91 + (9.09 \cdot 125.0/2630.183)/173.33$$

$$= 0.039$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load QO (Weight)	QO	26430.0 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	73.1 Kgf
Maximum Transverse Force	Ft	393.5 Kgf
Maximum Longitudinal Force	Fl	228.6 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 228.6 / (9.797 \cdot 4)$$

$$= 0.6 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 393.5 / (9.797 \cdot 4)$$

$$= 1.0 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 228.64 / (103.2 \cdot 4.0)$$

$$= 0.0543 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1300.0 \cdot 393.5 + 0.0$$

$$= 511.56 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 511.56/26429.97$$

$$= 19.35 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.0543, 0.0]$$

$$= 0.0543 \text{ cm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	866.50	mm.
Shell Thickness used in this Case	t	125.000	mm.
Stiffened Vessel Length per 4.15.6	L	5650.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	755.00	mm.

Saddle Width	b1 or b	250.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	765.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	461.75	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm.
Coefficient of Friction	mu	0.08	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		26711.06	Kgf
Pressure used in Analysis	P	255.288	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	90.15	234.00
Long. Stress at Bottom of Midspan	91.18	234.00
Long. Stress at Top of Saddles	91.71	234.00
Long. Stress at Bottom of Saddles	90.09	234.00
Tangential Shear in Shell	1.92	138.58
Circ. Stress at Horn of Saddle	1.63	350.99
Circ. Compressive Stress in Shell	0.19	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (787.0/2 + 0) \cdot 1300.0/1550.0$
 $= 990.1 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(228.64, 0) \cdot 1300.0/4068.9207$
 $= 73.1 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 25721 + \max(73, 990, 0, 0)$
 $= 26711.1 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169 (0.0 + \pi/4 (1.855 \cdot 1.2)^2)$
 $= 6794.525 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	27415.63	Kgf
Transverse Shear Load	393.50	Kgf
Longitudinal Shear Load	228.64	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 1550.000 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0427 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
 K7p = 0.0351

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -26711 \cdot 755.0 [1 - (1 - 755.0/5650.8 + (866.5^2 - 461.75^2)/(2 \cdot 755.0 \cdot 5650.8))/(1 + (4 \cdot 461.75)/(3 \cdot 5650.8))]$$

$$= -3265.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L]$$

$$= 26711 \cdot 5651/4 [(1 + 2(867^2 - 462^2)/(5651^2))/(1 + (4 \cdot 462)/(3 \cdot 5651)) - 4 \cdot 755/5651]$$

$$= 15006.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 255.288 \cdot 866.5/(2 \cdot 122.0) - 15006.6/(\pi \cdot 866.5^2 \cdot 122.0)$$

$$= 90.15 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 255.288 \cdot 866.5/(2 \cdot 122.0) + 15006.6/(\pi \cdot 866.5^2 \cdot 122.0)$$

$$= 91.18 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= 255.288 \cdot 866.5/(2 \cdot 122.0) - 3265.4/(0.1066 \cdot \pi \cdot 866.5^2 \cdot 122.0)$$

$$= 91.71 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 255.288 \cdot 866.5/(2 \cdot 122.0) + 3265.4/(0.1923 \cdot \pi \cdot 866.5^2 \cdot 122.0)$$

$$= 90.09 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 26711(5650.8 - 2 \cdot 755.0)/(5650.8 + (4 \cdot 461.75/3))$$

$$= 17650.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 17650.32/(866.5 \cdot 122.0)$$

$$= 1.92 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(866.5 \cdot 122.0)}$$

$$= 253.606 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(866.5 \cdot 122.0)}, 2 \cdot 755.0)$$

$$= 757.21 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 26711 \cdot 0.1/(757.21(122.0 + 1.0 \cdot 20.0))$$

$$= -0.19 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r^*]:

$$\begin{aligned}
 &= -Q/(4(t+\eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm/(L(t+\eta \cdot tr)^2) \\
 &= -26711/(4(122.0+ 1.0 \cdot 20.0)757.211) - \\
 &\quad 12 \cdot 0.043 \cdot 26711 \cdot 866.5/(5650.8(122.0+1.0 \cdot 20.0)^2) \\
 &= -1.63 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1550.0000	mm.
Baseplate Thickness	Bpthk	52.0000	mm.
Baseplate Width	Bpwid	275.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm.
Web Thickness	Webtk	20.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	500.0	mm.
Friction Coefficient	mu	0.080	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12 \\
 &= ((1.56 \cdot \sqrt{(805.5 \cdot 122.0)} + 765.0)122.0^3)/12 \\
 &= 18976.078 \text{ cm}^4
 \end{aligned}$$

Resolved Inertia for the Shell [I,shell]:

$$\begin{aligned}
 &= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2 \\
 &= 18976.078 + 1529.919(105.026 - 61.0)^2 \\
 &= 48630.113 \text{ cm}^4
 \end{aligned}$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1254.0	122.0	61.0	1529.9	9332505.0	0.486E+05
Wearplate	765.0	20.0	132.0	153.0	2019599.8	0.116E+04
Web	20.0	300.5	292.2	60.1	1756421.9	0.256E+05
BasePlate	275.0	52.0	468.5	143.0	6699548.5	0.189E+06
Totals	...	...	...	1886.0	19808076.0	0.265E+06

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 19808076/1886.019 \\
 &= 105.026 \text{ mm.}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120.0/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2)/(\pi - \beta + \sin(\beta)\cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2)/(\pi - 2.094 + \sin(120.0)\cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.204 \cdot 26711.057 \\
 &= 5436.2788 \text{ Kgf}
 \end{aligned}$$

Tension Stress, St = (Fh/As) = 1.4971 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2)/(\theta/2 \text{ in radians}) \\
 &= 1300.0 - 805.5 \cdot \sin(120.0/2)/1.0472 \\
 &= 633.857 \text{ mm.}
 \end{aligned}$$

Bending Moment, M = Fh * d = 3445.8906 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 1.3412 N./mm²

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(26711 + 705) \cdot 275.0}{4 \cdot 1550.0 \cdot 173.33}}$$

$$= 14.367 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (802.5 + 125.0 + 20.0)$$

$$= 1641.118 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 1641.118 / (5 - 1)$$

$$= 410.280 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 410.2795 \cdot 275.0 / 2$$

$$= 564.134 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 27415.633 / (1550.0 \cdot 275.0)$$

$$= 6.432 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 564.1 \cdot 6.43$$

$$= 3628.410 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 46.0 + 41.028$$

$$= 87.028 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 3628.4 / 87.0279$$

$$= 4.089 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	250.0	...	50.0	...	0.260E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 250.0 - 20.0$$

$$= 230.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 87.028$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0 / 2$$

$$= 125.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{2604.2 / 87.028}$$

= 54.702 mm.

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 1300.0 - cos(120.0/2)(802.5 + 125.0 + 20.0) - 52.0
 = 774.250 mm.

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
 = 2 • sin(120.0/2)(802.5 + 125.0 + 20.0)
 = 1641.118 mm.

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance
 = 1550.0 - 50.8
 = 1499.200 mm.

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2
 = (1641.118 - 1499.2)/2
 = 70.959 mm.

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$
 = $\sqrt{774.25^2 + 70.959^2}$
 = 777.495 mm.

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0}$
 = 123.211

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 777.495/54.702
 = 14.213

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 228.6 / (2 • 1550.0) • 410.28 • 777.49/2
 = 11.764 Kg-m.

Compressive Allowable, KL/r < Cc (14.2132 < 123.2106) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (14.21)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (14.21) / (8 \cdot 123.21) - (14.21^3) / (8 \cdot 123.21^3))$
 = 151.1 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
 = 4.09/151.06 + (11.76 • 125.0/26041666)/173.33
 = 0.030

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	230.0	0.0	46.0	0.0	0.260E+04
Web	410.3	20.0	0.0	82.1	0.0	27.4
Totals	...	...	...	128.1	...	0.263E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/128.056
 = 0.000 mm.

Distance to Centroid [C1]:

= Saddle Width / 2
 = 250.0/2

= 125.000 mm.

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 1300.0 - cos(947.5/2)(52.0 + 0.0 + 0.0) - 0.0
 = 300.500 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{2630.2/128.056}$
 = 45.320 mm.

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 300.5/45.32
 = 6.631

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 228.644/(2 • 1550.0)
 = 0.074 Kg/mm.

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 0.074 • 410.28 • 300.5
 = 9.093 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 9.093/210414.812
 = 0.424 N./mm²

Compressive Allowable, KL/r < Cc (6.6306 < 123.2106) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (6.63)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (6.63) / (8 \cdot 123.21) - (6.63^3) / (8 \cdot 123.21^3))$
 = 153.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 5.56/153.91 + (9.09 • 125.0/2630.183)/173.33
 = 0.039

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	9.7967 cm ²
Saddle Load Q0 (Weight)	Q0	26425.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	73.1 Kg
Maximum Transverse Force	Ft	393.5 Kg
Maximum Longitudinal Force	Fl	228.6 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= FI / (Bolt Area • Number of Bolts)
 = 228.6/(9.797 • 4)
 = 0.6 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= Ft / (Bolt Area • Number of Bolts)
 = 393.5/(9.797 • 4)
 = 1.0 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 228.64/(103.2 • 4.0)
= 0.0543 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 1300.0 • 393.5 + 0.0
= 511.56 Kg-m.

Eccentricity (e):
= Rmom / QO
= 511.56/26425.54
= 19.36 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.0543, 0.0]
= 0.0543 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: T2-Tube outlet

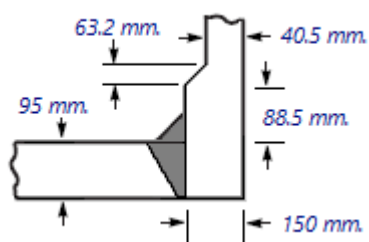
From: 30

Pressure for Reinforcement Calculations	P	147.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1605.00	mm.
Shell Finished (Minimum) Thickness	t	95.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1983.18	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	88.4530	mm.
Height of Beveled Transition	L'	63.2270	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		90.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: T2-Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 12.812 in.
 Actual Thickness Used in Calculation 1.594 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147.0 \cdot 805.5 / (138 \cdot 1.0 - 0.6 \cdot 147.0)$
 $= 91.6694$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147.0 \cdot 165.71 / (138 \cdot 1.0 - 0.6 \cdot 147.0)$
 $= 18.8587$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 109.512 / 0.5773$
 $= 189.6811$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

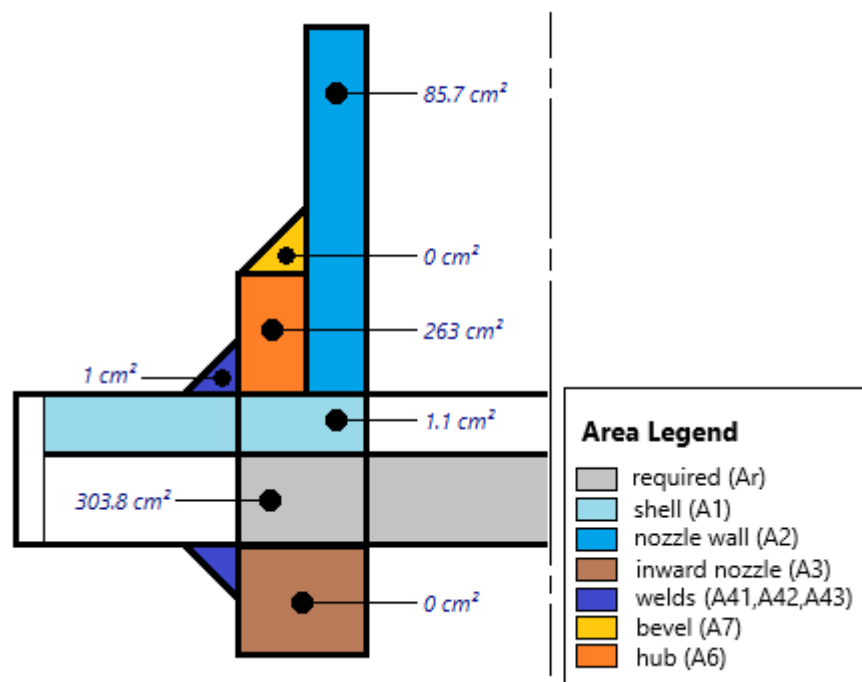
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	662.8496	mm.
Parallel to Vessel Wall, opening length	d	331.4248	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	230.0000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	303.815	NA	NA
Area in Shell	A1	1.096	NA	NA
Area in Nozzle Wall	A2	85.693	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	262.975	NA	NA
TOTAL AREA AVAILABLE	Atot	350.764	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (331.4248 \cdot 91.6694 \cdot 1.0 + 2 \cdot 147.0 \cdot 91.6694 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 303.815 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 331.425(1.0 \cdot 92.0 - 1.0 \cdot 91.669) - 2 \cdot 147.0$$

$$(1.0 \cdot 92.0 - 1.0 \cdot 91.6694) \cdot (1 - 1.0)$$

$$= 1.096 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 230.0)(37.49 - 18.86)1.0$$

$$= 85.693 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr_2$$

$$= (2 \cdot \min(230.0, 300.0, 88.5)) \cdot (150.0 - 40.5) \cdot 1.0$$

$$= 262.975 \text{ cm}^2 \text{ Includes Bevel Area of } 69.241 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 21.8587 \text{ mm.}$
 Wall Thickness per UG16(b), $tr_{16b} = 5.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb_1 = 94.6694 \text{ mm.}$
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 94.6694 \text{ mm.}$
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb_3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[11.331, \max(94.6694, 5.5)]$
 $= 11.3312 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(21.8587, 11.3312)$
 $= 21.8587 \text{ mm.}$

Available Nozzle Neck Thickness = $0.875 \cdot 40.488 = 35.427 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.3,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	318.7 N./mm ²	Passed
Occasional	:	29.2,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	8.2,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $tg = 35.427$, $tr = 18.859$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.582$, Temp. Reduction = $24 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	28 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	4 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $tg = 95.0$, $tr = 91.669$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.996$, Temp. Reduction = $0 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	30 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	30 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : 30 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-42 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $147.00 / 255.10 = 0.576$

Weld Size Calculations, Description: T2-Tube outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm.
---	------	-------------

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------

Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (303.8151 - 1.0957 + 2 \cdot 147.0 \cdot 1.0 \cdot (1.0 \cdot 92.0 - 91.6694)) \cdot 138)$
 $= 427340.19 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (348.6682 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 492036.62 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (348.6682 + 0.0 + 1. + (270.48)) \cdot 138$
 $= 872643.38 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (348.6682 + 0.0 + 1. + 0.0 + (270.48)) \cdot 138$
 $= 872643.38 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 625.4248 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 67738. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 239.2124) \cdot (150.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 1088157. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 625.4248 \cdot (90.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 920686. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (67738 + 1088157) = 1155895 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (67738 + 0 + 920687 + 0) = 988424 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (67738 + 920687 + 0) = 988424 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 1155894 Kgf, must exceed W = 427340 Kgf or W1 = 492036 Kgf
 Path 2-2 = 988424 Kgf, must exceed W = 427340 Kgf or W2 = 872643 Kgf
 Path 3-3 = 988424 Kgf, must exceed W = 427340 Kgf or W3 = 872643 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 147 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 63.4349 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 458.4349 mm.

Input Echo, WRC297 Item 1, Description: T2-Tube outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1605.000	mm.
Shell Thickness	95.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	325.425	mm.
Nozzle Thickness	150.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	147.00	bars
Radial Load	P	-13053.20	Kgf
Circumferential Shear	Vc	13053.20	Kgf
Longitudinal Shear	Vl	9789.91	Kgf
Circumferential Moment	Mc	13.57	Kg-m.
Longitudinal Moment	Ml	13.57	Kg-m.
Torsional Moment	Mt	15.66	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (18.895) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (4.169) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 625.425 + 2 \cdot 1.65 \cdot \sqrt{(851.5(95.0 - 3.0))}$
 $= 1549.059 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1703.000	mm.
Nozzle Outside Diameter	(d) =	625.425	mm.
Vessel Thickness used	(T) =	92.000	mm.
Nozzle Thickness used	(t) =	147.000	mm.
T / t	=	0.626	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.580$$

Nr/P = 0.084
 Mr/P = 0.160
 MØ/P = 0.048
 NØ/P = 0.109
 MrD/Mc = 0.286
 NrDL/Mc = 0.110
 MØd/Mc = 0.093
 NØDL/Mc = 0.058
 MrD/Ml = 0.139
 NrDL/Ml = 0.078
 MØD/Ml = 0.046
 NØDL/Ml = 0.132

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	121	136	121	136
Outplane Stress Summary	127	133	127	133

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	60	60	60	60
Inplane Stress Summary	76	47	76	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	127	133	127	133

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	60	60	60	60

Outplane Stress Summary	66	58	66	58
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	121	136	121	136

Inplane Stress Summary	137	123	137	123
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-1	-1	1	1
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	137	123	137	123
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	31	13	31

Hoop Stress Summary	15	33	15	33
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	11	3	11	3
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	15	33	15	33

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	31	13	31
Hoop Stress Summary	15	33	15	33

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	11	3	11	3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	15	33	15	33

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	121	136	121	136	121	136	121	136



Circ. P1 (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	4	-4	4	-4	14	-14	14	-14
Long. Pm (SUS)	60	60	60	60	60	60	60	60
Long. P1 (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	14	-14	14	-14	4	-4	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	121.0	136.0	121.0	136.0	121.0	136.0	121.0	136.0
Pm+P1 (SUS)	122.0	137.0	122.0	137.0	122.0	137.0	122.0	137.0
Pm+P1+Q (Total)	126.0	133.0	126.0	133.0	136.0	123.0	136.0	123.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	138.00	Passed
Pm+P1 (SUS)	137.01	207.00	Passed
Pm+P1+Q (TOTAL)	136.01	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	31	13	31	13	31	13	31
Circ. P1 (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. P1 (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	4	-4	4	-4	4	-4	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	13.0	31.0	13.0	31.0	13.0	31.0	13.0	31.0
Pm+P1 (SUS)	14.0	32.0	14.0	32.0	14.0	32.0	14.0	32.0
Pm+P1+Q (Total)	14.0	32.0	14.0	32.0	14.0	32.0	14.0	32.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	31.00	138.00	Passed
Pm+P1 (SUS)	32.00	207.00	Passed
Pm+P1+Q (TOTAL)	32.00	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: T1-Tube Inlet

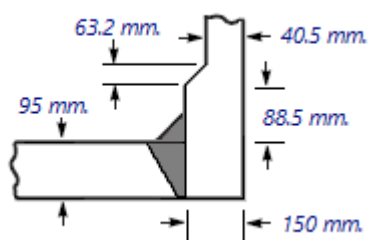
From: 30

Pressure for Reinforcement Calculations	P	147.157	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1605.00	mm.
Shell Finished (Minimum) Thickness	t	95.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1983.18	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	88.4530	mm.
Height of Beveled Transition	L'	63.2270	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		90.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: T1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 12.812 in.
 Actual Thickness Used in Calculation 1.594 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147.16 \cdot 805.5 / (138 \cdot 1.0 - 0.6 \cdot 147.16)$
 $= 91.7742 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147.16 \cdot 165.71 / (138 \cdot 1.0 - 0.6 \cdot 147.16)$
 $= 18.8803 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 109.512 / 0.5773$
 $= 189.6811 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

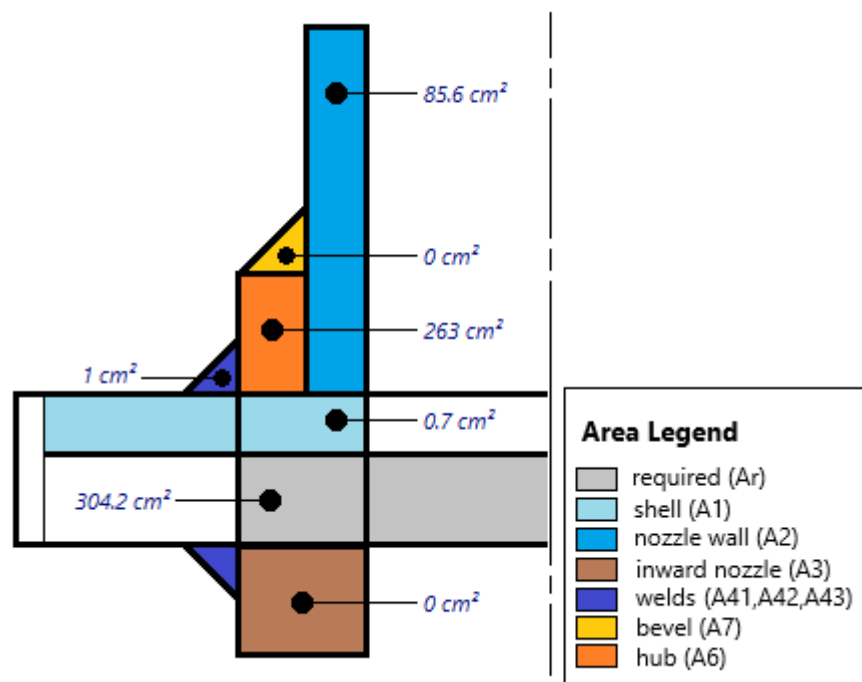
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	662.8496	mm.
Parallel to Vessel Wall, opening length	d	331.4248	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	230.0000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	304.163	NA	NA
Area in Shell	A1	0.748	NA	NA
Area in Nozzle Wall	A2	85.594	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	262.975	NA	NA
TOTAL AREA AVAILABLE	Atot	350.317	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (331.4248 \cdot 91.7742 \cdot 1.0 + 2 \cdot 147.0 \cdot 91.7742 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 304.163 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 331.425 (1.0 \cdot 92.0 - 1.0 \cdot 91.774) - 2 \cdot 147.0$$

$$(1.0 \cdot 92.0 - 1.0 \cdot 91.7742) \cdot (1 - 1.0)$$

$$= 0.748 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 230.0) (37.49 - 18.88) 1.0$$

$$= 85.594 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$



Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$
 $= (2 \cdot \min(230.0, 300.0, 88.5)) \cdot (150.0 - 40.5) \cdot 1.0$
 $= 262.975 \text{ cm}^2$ Includes Bevel Area of 69.241 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 21.8803 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 5.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 94.7742 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 94.7742 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 5.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.331, \max(94.7742, 5.5)]$
 $= 11.3312 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(21.8803, 11.3312)$
 $= 21.8803 \text{ mm.}$

Available Nozzle Neck Thickness = $0.875 \cdot 40.488 = 35.427 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.3,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	318.7 N./mm ²	Passed
Occasional	:	29.2,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	8.2,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $tg = 35.427$, $tr = 18.88$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.582$, Temp. Reduction = $24 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	28 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	4 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 95.0$, $tr = 91.774$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.998$, Temp. Reduction = $0 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	30 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	30 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : 30 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-42 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $147.16/255.10 = 0.577$

Weld Size Calculations, Description: T1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (304.1625 - 0.7483 + 2 \cdot 147.0 \cdot 1.0 \cdot (1.0 \cdot 92.0 - 91.7742)) \cdot 138)$
 $= 427884.22 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (348.5691 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 491897.03 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (348.5691 + 0.0 + 1. + (270.48)) \cdot 138$
 $= 872503.75 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (348.5691 + 0.0 + 1. + 0.0 + (270.48)) \cdot 138$
 $= 872503.75 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 625.4248 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 67738. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 239.2124) \cdot (150.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 1088157. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 625.4248 \cdot (90.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 920686. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (67738 + 1088157) = 1155895 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (67738 + 0 + 920687 + 0) = 988424 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (67738 + 920687 + 0) = 988424 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 1155894 Kgf, must exceed W = 427884 Kgf	or W1 = 491897 Kgf
Path 2-2 = 988424 Kgf, must exceed W = 427884 Kgf	or W2 = 872503 Kgf
Path 3-3 = 988424 Kgf, must exceed W = 427884 Kgf	or W3 = 872503 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 147 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 63.4349 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 458.4349 mm.

Input Echo, WRC297 Item 1, Description: T1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1605.000	mm.
Shell Thickness	95.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	325.425	mm.
Nozzle Thickness	150.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	147.16	bars
Radial Load	P	-13053.20	Kgf
Circumferential Shear	Vc	13053.20	Kgf
Longitudinal Shear	Vl	9789.91	Kgf
Circumferential Moment	Mc	13.57	Kg-m.
Longitudinal Moment	Ml	13.57	Kg-m.
Torsional Moment	Mt	15.66	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (18.895) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (4.169) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 625.425 + 2 \cdot 1.65 \cdot \sqrt{(851.5(95.0 - 3.0))}$
 $= 1549.059 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1703.000	mm.
Nozzle Outside Diameter	(d) =	625.425	mm.
Vessel Thickness used	(T) =	92.000	mm.
Nozzle Thickness used	(t) =	147.000	mm.
T / t	=	0.626	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.580$$



Nr/P	=	0.084
Mr/P	=	0.160
M0/P	=	0.048
N0/P	=	0.109
MrD/Mc	=	0.286
NrDL/Mc	=	0.110
M0d/Mc	=	0.093
N0DL/Mc	=	0.058
MrD/Ml	=	0.139
NrDL/Ml	=	0.078
M0D/Ml	=	0.046
N0DL/Ml	=	0.132

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	121	136	121	136
Outplane Stress Summary	127	133	127	133

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	60	60	60	60
Inplane Stress Summary	76	47	76	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	127	133	127	133

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	60	60	60	60
Outplane Stress Summary	66	58	66	58

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	121	136	121	136
Inplane Stress Summary	137	123	137	123

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	137	123	137	123

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	31	13	31
Hoop Stress Summary	15	33	15	33

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	11	3	11	3

Nozzle Stresses (N./mm²):



LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	15	33	15	33

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	31	13	31
Hoop Stress Summary	15	33	15	33

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	11	3	11	3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	15	33	15	33

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	121	136	121	136	121	136	121	136



Circ. P1 (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	4	-4	4	-4	14	-14	14	-14
Long. Pm (SUS)	60	60	60	60	60	60	60	60
Long. P1 (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	14	-14	14	-14	4	-4	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	121.0	136.0	121.0	136.0	121.0	136.0	121.0	136.0
Pm+P1 (SUS)	122.0	137.0	122.0	137.0	122.0	137.0	122.0	137.0
Pm+P1+Q (Total)	126.0	133.0	126.0	133.0	136.0	123.0	136.0	123.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	138.00	Passed
Pm+P1 (SUS)	137.01	207.00	Passed
Pm+P1+Q (TOTAL)	136.01	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	31	13	31	13	31	13	31
Circ. P1 (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. P1 (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	4	-4	4	-4	4	-4	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	13.0	31.0	13.0	31.0	13.0	31.0	13.0	31.0
Pm+P1 (SUS)	14.0	32.0	14.0	32.0	14.0	32.0	14.0	32.0
Pm+P1+Q (Total)	14.0	32.0	14.0	32.0	14.0	32.0	14.0	32.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	31.00	138.00	Passed
Pm+P1 (SUS)	32.00	207.00	Passed
Pm+P1+Q (TOTAL)	32.00	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

Input, Nozzle Desc: S1-Shell Inlet

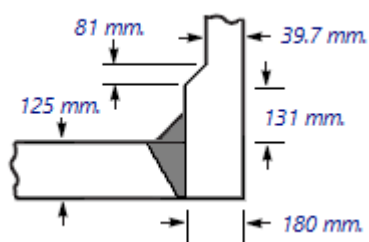
From: 40

Pressure for Reinforcement Calculations	P	191.000	bars
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1605.00	mm.
Shell Finished (Minimum) Thickness	t	125.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4373.17	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		140	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	131.0170	mm.
Height of Beveled Transition	L'	81.0168	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		118.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: S1-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 14.876 in.
 Actual Thickness Used in Calculation 1.562 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 191.0 \cdot 805.5 / (138 \cdot 1.0 - 0.6 \cdot 191.0)$
 $= 121.5930$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 191.0 \cdot 191.93 / (138 \cdot 1.0 - 0.6 \cdot 191.0)$
 $= 28.9717$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 140.325 / 0.5773$
 $= 243.0505$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

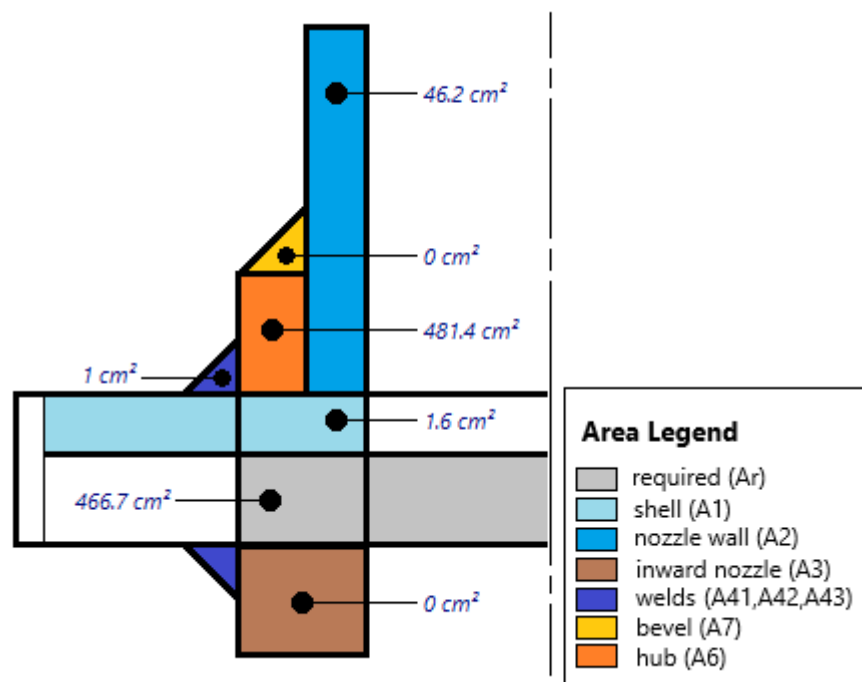
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	767.7007	mm.
Parallel to Vessel Wall, opening length	d	383.8504	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	303.7208	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	466.735	NA	NA
Area in Shell	A1	1.562	NA	NA
Area in Nozzle Wall	A2	46.219	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	481.387	NA	NA
TOTAL AREA AVAILABLE	Atot	530.168	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (383.8504 \cdot 121.593 \cdot 1.0 + 2 \cdot 177.0 \cdot 121.593 \cdot 1.0 \cdot (1 - 1.0))$
 $= 466.735 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 383.85 (1.0 \cdot 122.0 - 1.0 \cdot 121.593) - 2 \cdot 177.0$
 $(1.0 \cdot 122.0 - 1.0 \cdot 121.593) \cdot (1 - 1.0)$
 $= 1.562 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot ho) (tn - trn) fr2$
 $= (2 \cdot 300.0) (36.67 - 28.97) 1.0$
 $= 46.219 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$
 $= (2 \cdot \min(300.0, 300.0, 131.0)) \cdot (180.0 - 39.7) \cdot 1.0$
 $= 481.387 \text{ cm}^2$ Includes Bevel Area of 113.687 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 31.9717 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 124.5930 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 124.5930 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.331, \max(124.593, 4.5)]$
 $= 11.3312 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(31.9717, 11.3312)$
 $= 31.9717 \text{ mm.}$

Available Nozzle Neck Thickness = $0.875 \cdot 39.675 = 34.715 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	42.8,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	302.2 N./mm ²	Passed
Occasional	:	45.6,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	8.2,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $tg = 34.715$, $tr = 28.972$, $c = 3.0 \text{ mm.}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.913$, Temp. Reduction = $5 \text{ }^{\circ}\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	28 $^{\circ}\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	23 $^{\circ}\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : 23 $^{\circ}\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^{\circ}\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-32 $^{\circ}\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $191.00 / 255.10 = 0.749$

Weld Size Calculations, Description: S1-Shell Inlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (466.7354 - 1.5621 + 2 \cdot 177.0 \cdot 1.0 \cdot (1.0 \cdot 122.0 - 121.593)) \cdot 138)$$

$$= 656597.00 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (527.6055 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 743828.62 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (527.6055 + 0.0 + 1. + (431.88)) \cdot 138$$

$$= 1351549.88 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (527.6055 + 0.0 + 1. + 0.0 + (431.88)) \cdot 138$$

$$= 1351549.88 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 737.8504 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 79914. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 280.4252) \cdot (180.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 1535963. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 737.8504 \cdot (118.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 1424113. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (79914 + 1535963) = 1615878 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (79914 + 0 + 1424113 + 0) = 1504028 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (79914 + 1424113 + 0) = 1504028 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1615877 Kgf, must exceed W = 656597 Kgf or W1 = 743828 Kgf
 Path 2-2 = 1504027 Kgf, must exceed W = 656597 Kgf or W2 = 1351549 Kgf
 Path 3-3 = 1504027 Kgf, must exceed W = 656597 Kgf or W3 = 1351549 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 192 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.



The Drop for this Nozzle is : 89.8287 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 514.8287 mm.

Input Echo, WRC297 Item 1, Description: S1-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1605.000	mm.
Shell Thickness	125.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	377.850	mm.
Nozzle Thickness	180.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	191.00	bars
Radial Load	P	-14684.89	Kgf
Circumferential Shear	Vc	14684.89	Kgf
Longitudinal Shear	Vl	11013.68	Kgf
Circumferential Moment	Mc	17.17	Kg-m.
Longitudinal Moment	Ml	47.76	Kg-m.
Torsional Moment	Mt	16.76	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (14.840) is < 20
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (4.099) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 737.85 + 2 \cdot 1.65 \cdot \sqrt{(866.5(125.0 - 3.0))}$
 $= 1810.797 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1733.000 mm.
Nozzle Outside Diameter	(d) =	737.850 mm.
Vessel Thickness used	(T) =	122.000 mm.
Nozzle Thickness used	(t) =	177.000 mm.
T / t	=	0.689
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.605$$

$$Nr/P = 0.086$$



Mr/P = 0.154
 MØ/P = 0.047
 NØ/P = 0.108
 MrD/Mc = 0.281
 NrDL/Mc = 0.116
 MØd/Mc = 0.093
 NØDL/Mc = 0.057
 MrD/Ml = 0.137
 NrDL/Ml = 0.079
 MØD/Ml = 0.047
 NØDL/Ml = 0.133

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	117	136	117	136
Outplane Stress Summary	121	134	121	134

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	58	58	58
Inplane Stress Summary	68	50	68	50

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	121	134	121	134

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58

Outplane Stress Summary	62	56	62	56
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	117	136	117	136

Inplane Stress Summary	126	128	127	128
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	126	128	127	128
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	39	17	39

Hoop Stress Summary	18	40	18	40
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	12	5	12	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Shear stress)	Outside	Inside	Outside	Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	18	40	18	40
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	39	17	39

Hoop Stress Summary	18	40	18	40
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	12	5	12	5
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	18	40	18	40
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	117	136	117	136	117	136	117	136
Circ. Pl (SUS)	1	1	1	1	0	0	0	0

Circ. Q (SUS)	2	-2	2	-2	8	-8	8	-8
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	0	0	0	0	1	1	1	1
Long. Q (SUS)	8	-8	8	-8	2	-2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	117.0	136.0	117.0	136.0	117.0	136.0	117.0	136.0
Pm+Pl (SUS)	118.0	137.0	118.0	137.0	117.0	136.0	117.0	136.0
Pm+Pl+Q (Total)	120.0	135.0	120.0	135.0	125.0	128.0	125.0	128.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	138.00	Passed
Pm+Pl (SUS)	137.01	207.00	Passed
Pm+Pl+Q (TOTAL)	135.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	17	39	17	39	17	39	17	39
Circ. Pl (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	3	-3	3	-3	3	-3	3	-3
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	17.0	39.0	17.0	39.0	17.0	39.0	17.0	39.0
Pm+Pl (SUS)	18.0	40.0	18.0	40.0	18.0	40.0	18.0	40.0
Pm+Pl+Q (Total)	18.0	40.0	18.0	40.0	18.0	40.0	18.0	40.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.00	138.00	Passed
Pm+Pl (SUS)	40.00	207.00	Passed
Pm+Pl+Q (TOTAL)	40.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: S2-Shell outlet

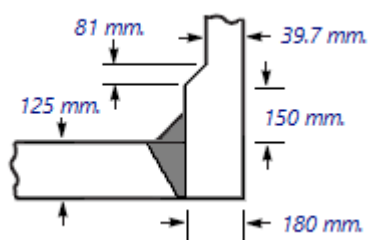
From: 40

Pressure for Reinforcement Calculations	P	191.008	bars
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1605.00	mm.
Shell Finished (Minimum) Thickness	t	125.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		6657.89	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		140	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	150.0000	mm.
Height of Beveled Transition	L'	81.0168	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		118.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: S2-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 14.876 in.
 Actual Thickness Used in Calculation 1.562 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 191.01 \cdot 805.5 / (138 \cdot 1.0 - 0.6 \cdot 191.01)$
 $= 121.5985 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 191.01 \cdot 191.93 / (138 \cdot 1.0 - 0.6 \cdot 191.01)$
 $= 28.9730 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 140.325 / 0.5773$
 $= 243.0505 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

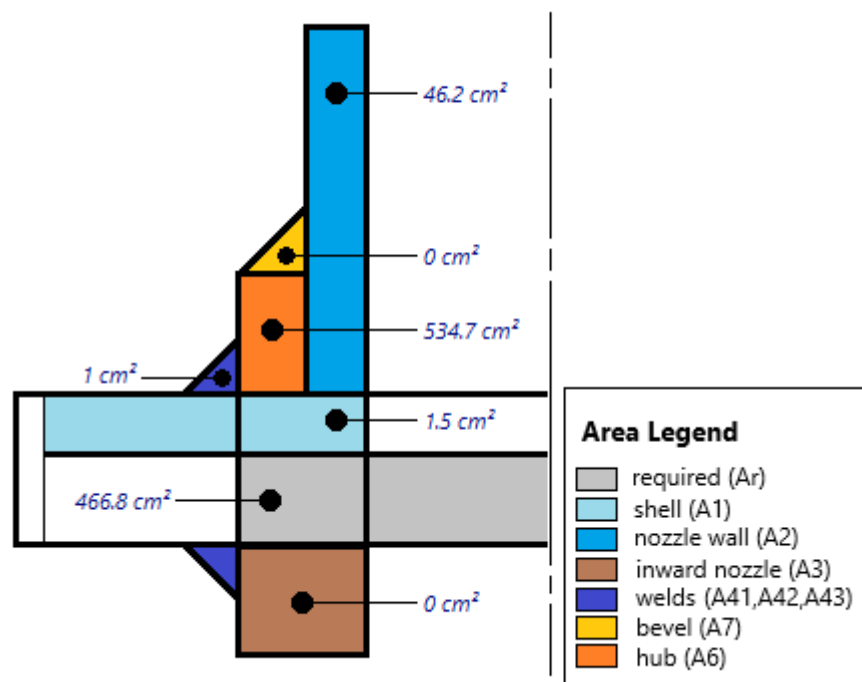
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	767.7007	mm.
Parallel to Vessel Wall, opening length	d	383.8504	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	305.0000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	466.756	NA	NA
Area in Shell	A1	1.541	NA	NA
Area in Nozzle Wall	A2	46.211	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	534.663	NA	NA
TOTAL AREA AVAILABLE	Atot	583.415	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (383.8504 \cdot 121.5985 \cdot 1.0 + 2 \cdot 177.0 \cdot 121.5985 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 466.756 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 383.85 (1.0 \cdot 122.0 - 1.0 \cdot 121.599) - 2 \cdot 177.0$$

$$(1.0 \cdot 122.0 - 1.0 \cdot 121.5985) \cdot (1 - 1.0)$$

$$= 1.541 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot ho) (tn - trn) fr2$$

$$= (2 \cdot 300.0) (36.67 - 28.97) 1.0$$

$$= 46.211 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$
 $= (2 \cdot \min(300.0, 300.0, 150.0)) \cdot (180.0 - 39.7) \cdot 1.0$
 $= 534.663 \text{ cm}^2$ Includes Bevel Area of 113.687 cm^2

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 31.9730 \text{ mm.}$
 Wall Thickness per UG16(b), $tr_{16b} = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 124.5985 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr_{16b}) = 124.5985 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr_{16b}) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.331, \max(124.5985, 4.5)]$
 $= 11.3312 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(31.973, 11.3312)$
 $= 31.9730 \text{ mm.}$

Available Nozzle Neck Thickness $= 0.875 \cdot 39.675 = 34.715 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	42.7,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	302.3 N./mm ²	Passed
Occasional	:	45.6,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	8.2,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 34.715$, $t_r = 28.973$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.914$, Temp. Reduction $= 5 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	28 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	23 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : 23 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-32 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating $= 191.01/255.10 = 0.749$

Weld Size Calculations, Description: S2-Shell outlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$



Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (466.7564 - 1.5411 + 2 \cdot 177.0 \cdot 1.0 \cdot (1.0 \cdot 122.0 - 121.5985)) \cdot 138)$$

$$= 656628.88 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (580.8735 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 818784.88 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (580.8735 + 0.0 + 1. + (431.88)) \cdot 138$$

$$= 1426506.12 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (580.8735 + 0.0 + 1. + 0.0 + (431.88)) \cdot 138$$

$$= 1426506.12 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 737.8504 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 79914. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 280.4252) \cdot (180.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 1535963. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 737.8504 \cdot (118.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 1424113. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (79914 + 1535963) = 1615878 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (79914 + 0 + 1424113 + 0) = 1504028 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (79914 + 1424113 + 0) = 1504028 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1615877 Kgf, must exceed W = 656628 Kgf or W1 = 818784 Kgf
 Path 2-2 = 1504027 Kgf, must exceed W = 656628 Kgf or W2 = 1426506 Kgf
 Path 3-3 = 1504027 Kgf, must exceed W = 656628 Kgf or W3 = 1426506 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 192 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 89.8287 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 514.8287 mm.

Input Echo, WRC297 Item 1, Description: S2-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1605.000	mm.
Shell Thickness	125.0000	mm.
Shell Stress Concentration Factor	1.200	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	377.850	mm.
Nozzle Thickness	180.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.*

Design Internal Pressure	Dp	191.01	bars
Radial Load	P	-14684.89	Kgf
Circumferential Shear	Vc	14684.89	Kgf
Longitudinal Shear	Vl	11013.68	Kgf
Circumferential Moment	Mc	17.17	Kg-m.
Longitudinal Moment	Ml	17.17	Kg-m.
Torsional Moment	Mt	19.82	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (14.840) is < 20
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (4.099) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 737.85 + 2 \cdot 1.65 \cdot \sqrt{(866.5(125.0 - 3.0))}$
 $= 1810.797 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1733.000 mm.
Nozzle Outside Diameter	(d) =	737.850 mm.
Vessel Thickness used	(T) =	122.000 mm.
Nozzle Thickness used	(t) =	177.000 mm.
T / t	=	0.689
d / T	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.605$$

$$Nr/P = 0.086$$



Mr/P = 0.154
 MØ/P = 0.047
 NØ/P = 0.108
 MrD/Mc = 0.281
 NrDL/Mc = 0.116
 MØd/Mc = 0.093
 NØDL/Mc = 0.057
 MrD/Ml = 0.137
 NrDL/Ml = 0.079
 MØD/Ml = 0.047
 NØDL/Ml = 0.133

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	117	136	117	136
Outplane Stress Summary	121	134	121	134

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	58	58	58
Inplane Stress Summary	70	48	70	48

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	121	134	121	134

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58

Outplane Stress Summary	63	56	63	56
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	117	136	117	136

Inplane Stress Summary	128	126	128	126
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	128	126	128	126
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	39	17	39

Hoop Stress Summary	18	40	18	40
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	12	5	12	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----

(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	18	40	18	40

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	39	17	39
Hoop Stress Summary	18	40	18	40

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	12	5	12	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	18	40	18	40

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	117	136	117	136	117	136	117	136
Circ. Pl (SUS)	1	1	1	1	0	0	0	0

Circ. Q (SUS)	3	-3	3	-3	10	-10	10	-10
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	0	0	0	0	1	1	1	1
Long. Q (SUS)	10	-10	10	-10	3	-3	3	-3
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	117.0	136.0	117.0	136.0	117.0	136.0	117.0	136.0
Pm+Pl (SUS)	118.0	137.0	118.0	137.0	117.0	136.0	117.0	136.0
Pm+Pl+Q (Total)	121.0	134.0	121.0	134.0	127.0	126.0	127.0	126.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	138.00	Passed
Pm+Pl (SUS)	137.01	207.00	Passed
Pm+Pl+Q (TOTAL)	134.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	17	39	17	39	17	39	17	39
Circ. Pl (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	3	-3	3	-3	3	-3	3	-3
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	17.0	39.0	17.0	39.0	17.0	39.0	17.0	39.0
Pm+Pl (SUS)	18.0	40.0	18.0	40.0	18.0	40.0	18.0	40.0
Pm+Pl+Q (Total)	18.0	40.0	18.0	40.0	18.0	40.0	18.0	40.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.00	138.00	Passed
Pm+Pl (SUS)	40.00	207.00	Passed
Pm+Pl+Q (TOTAL)	40.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	191.00 bars
Shell Maximum Operating Pressure	Psox,max	191.00 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	125.0000 mm.
Shell Internal Corrosion Allowance	cas	3.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1605.000 mm.
Shell Temperature for Internal Pressure	Ts	200.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	147.00 bars
Channel Maximum Operating Pressure	Ptox,max	147.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	95.0000 mm.
Channel Internal Corrosion Allowance	cac	3.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1605.000 mm.
Channel Design Temperature	TEMPC	85.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1126
Tube Wall Thickness	et	3.4040 mm.
Tube Outside Diameter	D	25.4000 mm.
Total Straight Tube Length	Lt	5150.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		4860.00 mm.
Design Temperature of the Tubes		200.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	154.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	40.6400 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.4000 mm.
Groove Weld Leg	ag	3.4000 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	764.093 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	15688.0 cm ²
Perimeter of Tube Layout	Cp	4749.9 mm.

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 200.00 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

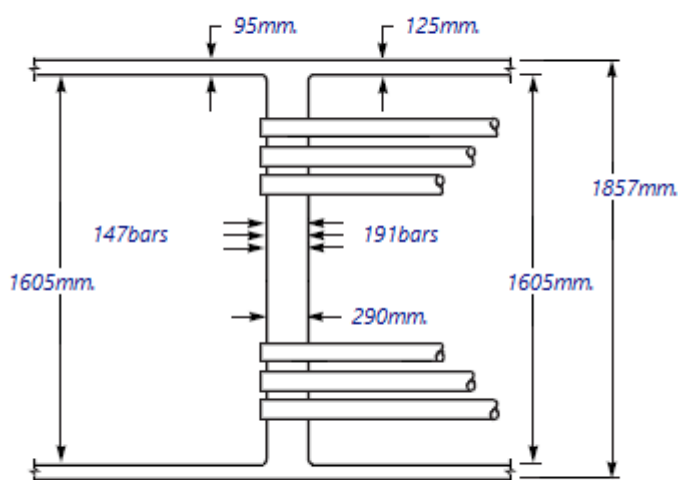
Tubesheet Material UNS Number K03506
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 290.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 3.0000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm.
 Tubesheet Outside Diameter A 1857.000 mm.

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	200.0 °C	0.19120E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Channel - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	85.0 °C	0.19766E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%



Elastic Mod. at Tubsht. Design Temp. 200.0 °C 0.19258E+09 KPa.
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 200.0 °C 0.19258E+09 KPa.
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (147.0 \cdot (25.4/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 147.0)$
 $= 1.8997 + 0.0000 = 1.8997 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 200.00 °C
 Elastic Modulus for Material 193602528.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
3.4040	25.40	5624.09	7.46	50.0000	0.0197562	93.30

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 193.2327 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
3.3762	25.40	5624.09	7.52	50.0000	0.0194347	93.30

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 191.0185 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	3.3762 mm.
Allowable Internal Pressure at Corroded thickness	277.38 bars
Required Internal Design Pressure	147.00 bars
Allowable External Pressure at Corroded thickness	193.23 bars
Required External Design Pressure	191.00 bars
Required Thickness due to Shell Side pressure	3.3762 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot ro + dt$
 $= 2 \cdot 764.093 + 25.4$
 $= 1553.586 \text{ mm.}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (40.64 - 25.4) / 40.64$
 $= 0.375$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= Ds / Do$ (Configurations a, b, c)
 $= 1605.0 / 1553.5861$
 $= 1.0331$

Ratio [pc]:

$= Dc / Do$ (Configurations a, e, f)
 $= 1605.0 / 1553.5861$
 $= 1.0331$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio $[\rho]$:

$$= l_{tx} / h$$

$$= 0/290.0$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter $[d^*]$:

$$= \max(dt - 2t_t(E_t/E) \cdot (St_t/S) \cdot (\rho); dt - 2t_t)$$

$$= \max(25.4 - 2 \cdot 3.404 (0.19258E+09/.19258E+09) \cdot$$

$$(92/138) \cdot (0); 25.4 - 2 \cdot 3.404)$$

$$= 25.4000 \text{ mm.}$$

Effective tube pitch $[p^*]$:

$$= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 40.64 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1553.586 \cdot 40.64) / (\pi \cdot 1553.586^2)) }$$

$$= 40.6400 \text{ mm.}$$

Effective ligament efficiency for bending $[\mu^*]$:

$$= (p^* - d^*) / p^*$$

$$= (40.64 - 25.4) / 40.64$$

$$= 0.3750$$

$h/p = 7.135827$; $\mu^* = 0.375000$; $E^* = 73241832 \text{ KPa.}$

$$E^*/E = 0.380317$$
; $v^* = 0.317427$

Note: As h/p (7.136) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient $[\beta_s]$:

$$= ((12(1 - v_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1605.0 + 125.0) 125.0) }$$

$$= 0.0039 \text{ 1/mm.}$$

Shell coefficient $[k_s]$:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2))$$

$$= 0.004 \cdot 1912019 \cdot 125.0^3 / (6(1 - 0.3^2))$$

$$= 2673666304.0000 \text{ bars} \cdot \text{mm.}^2$$

Shell coefficient $[\lambda_s]$:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1605.0 + 125.0) \cdot 0.26737E+1 / 290.0^3 \cdot (1 + 290.0 \cdot 0.004 + (290.0 \cdot 0.004)^2 / 2)$$

$$= 3159098.8 \text{ bars}$$

Shell coefficient $[\delta_s]$:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1605.0^2 / (4 \cdot 0.19120E+09 \cdot 125.0) \cdot (1 - 1605.0 \cdot 0.3 / (2(1605.0 + 125.0)))$$

$$= 0.23194E-01 \text{ mm./N./mm}^2$$

Parameter $[\omega_s]$:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.0331 \cdot 0.26737E+1 \cdot 0.0039 \cdot 0.023194 (1 + 290.0 \cdot 0.0039)$$

$$= 53438.6211 \text{ mm.}^2$$

Channel coefficient $[\beta_c]$:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1605.0 + 95.0) 95.0) }$$

$$= 0.0045 \text{ 1/mm.}$$

Channel coefficient $[k_c]$:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 1976633 \cdot 95.0^3 / (6(1 - 0.3^2))$$

$$= 1404021376.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients $[CC, CqDelta, CqTheta, CmTheta, CmDelta]$:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient $[\lambda_c]$:



$$\begin{aligned}
 &= 6(Dc + tc/\cos(\theta)) \cdot kc/h^3 \cdot (Cm\theta + h \cdot \beta c \cdot (Cm\Delta + Cq\theta)/2 + (h \cdot \beta c)^2/2 \cdot Cq\Delta) \\
 &= 6(1605.0 + 95.0/\cos(0)) \cdot 0.14040E+1/290.0^3 \cdot (1.0 + 290.0 \cdot 0.005 \cdot (1.0 + 1.0)/2 + (290.0 \cdot 0.005)^2/2 \cdot 1.0) \\
 &= 1862694.1 \text{ bars}
 \end{aligned}$$

Channel coefficient [δc]:

$$\begin{aligned}
 &= Dc^2/(4 \cdot Ec \cdot tc) \cdot (1 - Dc \cdot vc/(2(Dc + tc))) \\
 &= 1605.0^2/(4 \cdot 0.19766E+09 \cdot 95.0) \cdot (1 - 1605.0 \cdot 0.3/(2(1605.0 + 95.0))) \\
 &= 0.29437E-01 \text{ mm./N./mm}^2
 \end{aligned}$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ωc]:

$$\begin{aligned}
 &= \rho c \cdot kc [\delta c (Cm\theta + \beta c + Cq\Delta \cdot h \cdot \beta c^2) - \zeta c (Cm\theta + h \cdot Cq\theta \cdot h \cdot \beta c / 2)] \\
 &= 1.033 \cdot 0.14040E+1 [0.029437 (1.0 + 0.005 \cdot 1.0 \cdot 290.0) - 0 (1.0/290.0 + 1.0 \cdot 0.005/2)] \\
 &= 44653.3516 \text{ mm.}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned}
 &= A / Do \\
 &= 1857.0/1553.5861 \\
 &= 1.1953
 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned}
 &= (1 - v^*)/E^* \cdot (\lambda s + \lambda c + E \cdot \ln(K)) \\
 &= (1 - 0.32)/73241832 \cdot (3159098 + 1862694 + 0.19258E+09 \cdot \ln(1.2)) \\
 &= 5.0002
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt] \\
 &= 1553.586^2/16 \cdot [(1.033 - 1) \cdot (1.033^2 + 1) \cdot 191.0 - (1.033 - 1) \cdot (1.033^2 + 1) \cdot 147.0] \\
 &= 454091.1250 \text{ bars}\cdot\text{mm.}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= Mts + \omega c \cdot Pt - \omega s \cdot Ps \\
 &= 454091.125 + 44653.352 \cdot 147.0 - 53438.621 \cdot 191.0 \\
 &= -3188614.5000 \text{ bars}\cdot\text{mm.}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned}
 &= ((M^*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F) \\
 &= ((-3188614.5) - 1553.586^2/32 \cdot 5.0 (191.0 - 147.0)) / (1 + 5.0) \\
 &= -3297019.0000 \text{ bars}\cdot\text{mm.}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= Mp + Do^2/64 \cdot (3 + v^*) \cdot (Ps - Pt) \\
 &= -3297019.0 + 1553.586^2/64 \cdot (3 + 0.317) \cdot (191.0 - 147.0) \\
 &= 2207752.0000 \text{ bars}\cdot\text{mm.}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(| Mp |; | Mo |) \\
 &= \max(| -3297019.0 |; | 2207752 |) \\
 &= 3297019.0000 \text{ bars}\cdot\text{mm.}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu^* (h - hg')^2) \\
 &= 6 \cdot 3297019 / (0.375 (290.0 - 0)^2) \\
 &= 62.7293 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138.0$$

$$= 276.00 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 3280998 / (0.375 (138.2543 - 0)^2)$$

$$= 274.6589 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$= h + C_{ats} + C_{atc}$$

$$= 138.2543 + 0 + 0$$

$$= 138.2543 \text{ mm.}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(138.2543 ; 35.1037)$$

$$= 138.2543 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| P_s - P_t |$]:

$$= | P_s - P_t |$$

$$= | 191.0 - 147.0 |$$

$$= 43.999 \text{ bars}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.375 \cdot 290.0 / 1553.59 \cdot \min(0.8 \cdot 138.0 ; 0.533 \cdot 213.0)$$

$$= 30.912 \text{ N./mm}^2$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.375 \cdot 110.4) \cdot (4 \cdot 15688.0 / 4749.92) \cdot | 191.0 - 147.0 | + 0$$

$$= 35.1037 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 125.000mm.

Min. Shell len. adjacent to the tubesheet is: 806.241 mm.

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1605.0^2 / (4 \cdot 125.0 (1605.0 + 125.0)) \cdot 191.0$$

$$= 56.8842 \text{ N./mm}^2$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P's + 6(1 - \nu) / E \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 60107784 / 125.0^2 \cdot [0.004 \cdot 0.4375 + 6(1 - 0.317) / 73241832 \cdot 1553.586 / 290.0^3 \cdot (1 + 290.0 \cdot 0.004 / 2) \cdot (-3297019.0 + 1553.586^2 / 32 \cdot (191.0 - 147.0))]$$

$$= 179.063 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s \text{)}$$

$$= 235.948 \text{ N./mm}^2 \text{ (Should be } \leq 207.00 \text{)}$$

Note: For a given Channel thickness of: 95.000 mm.

Min. Channel len. adjacent to the tubesheet is: 702.865mm.

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1605.0^2 / (4 \cdot 95.0 (1605.0 + 95.0) \cdot \cos(0)) \cdot 147.0$$

$$= 58.622 \text{ N./mm}^2$$



Axial Channel Bending Stress [σ_c, b]:

$$\begin{aligned}
 &= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \Theta) \cdot P_t - \\
 &\quad 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \Theta + \\
 &\quad h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))] \\
 &= 6 \cdot 31564380 / 95.0^2 \cdot [(0.029437 \cdot 1.0 \cdot 0.005 - 0 \cdot 1.0) \cdot 147.0 - \\
 &\quad 6((1 - 0.317) / 73241832) \cdot 1553.586 / 290.0^3 \cdot (1.0 + \\
 &\quad 0.005 \cdot 290.0 / 2 \cdot 1.0) \cdot (-3297019.0 + 1553.586^2 / 32 \cdot (191.0 - 147.0))] \\
 &= 181.537 \text{ N./mm}^2
 \end{aligned}$$

Channel Membrane + Bending Stress [σ_c]:

$$\begin{aligned}
 &= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c) \\
 &= 240.159 \text{ N./mm}^2 \text{ (Should be } \leq 207.00)
 \end{aligned}$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_s]:

$$\begin{aligned}
 &= E_s \cdot \sqrt{(1.5 \cdot S_s / \sigma_s)} \\
 &= 0.19120E+09 \cdot \sqrt{(1.5 \cdot 138.0 / 235)} \\
 &= 179087696.0 \text{ KPa.}
 \end{aligned}$$

Modified Elasticity [E_c]:

$$\begin{aligned}
 &= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)} \\
 &= 0.19766E+09 \cdot \sqrt{(1.5 \cdot 138.0 / 240.16)} \\
 &= 183509280.0 \text{ KPa.}
 \end{aligned}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - h'g)^2) \\
 &= 6 \cdot 3295834 / (0.375 \cdot (290.0 - 0)^2) \\
 &= 62.7 \text{ N./mm}^2 \text{ (Should be } \leq 276.0)
 \end{aligned}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:

$$= h + C_{ats} + C_{atc} = 138.2543 + 0 + 0 = 138.2543 \text{ mm.}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS}) = \max(138.2543; 35.1037) = 138.2543 \text{ mm.}$$

Tube Metal Area [A_t]:

$$\begin{aligned}
 &= \pi (d_o - t) t \\
 &= \pi (25.4 - 3.404) 3.404 \\
 &= 2.352 \text{ cm}^2
 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$\begin{aligned}
 &= \pi \cdot t (d_o - t) S \\
 &= \pi \cdot 3.404 (25.4 - 3.404) 92.398 \\
 &= 2216.250 \text{ Kgf}
 \end{aligned}$$

Groove Weld Strength [F_g]:

$$\begin{aligned}
 &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\
 &= 0.85 \cdot \pi \cdot 3.4 (25.4 + 0.67 \cdot 3.4) 92.4 \\
 &= 2216.2495 \text{ Kgf}
 \end{aligned}$$

Weld Strength Factor [f_w]:

$$\begin{aligned}
 &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\
 &= 92.398 / 92.398 \\
 &= 1.000
 \end{aligned}$$

Design Strength Ratio [f_d]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 3.4 (25.4 + 0.67 \cdot 3.4) 92.4$$

$$= 1532.0105 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 3.404 \cdot (25.4 - 3.404) 1.0 \cdot 1.0} - 0.75 \cdot 25.4 \right)$$

$$= 3.996 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 2.0424 mm.

Minimum Required Groove Weld Leg [agr]: 2.0424 mm.

Tube-Tubesheet Jt allowable, 2216.25 is >= tube strength 2216.25 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	62.7 <=	276.0 N./mm ²	Ok
Tubesheet shear stress	13.4 <=	110.4 N./mm ²	Ok
Stress in Shell at Tubesheet	235.9 <=	450.0 N./mm ²	Ok
Stress in Channel at Tubesheet	240.2 <=	484.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	138.254	290.000	Ok
Tube-Tubesheet Fillet Weld Leg :	2.042	3.400	Ok
Tube-Tubesheet Groove Weld Leg :	2.042	3.400	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D1uc	244.119	...	191	276	45	110	Ps+Pt D1	Ok
D2uc	278.838	...	253	276	58	110	Ps+Pt D2	Ok
D3uc	138.254	...	63	276	13	110	Ps+Pt D3	Ok
D1c	249.290	...	197	276	46	110	Ps+Pt-c D1	Ok
D2c	283.843	...	262	276	59	110	Ps+Pt-c D2	Ok
D3c	143.934	...	65	276	14	110	Ps+Pt-c D3	Ok
Max:	283.8431	...	mm.	0.948		0.537	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	Shell	Channel	Shell Stress	Shell Req'd. Thk.	Channel Stress	Channel Allwd	Pass Fail
D1uc	...	...	58	207	297	484	Ok
D2uc	...	...	294	450	57	207	Ok
D3uc	...	...	236	450	240	484	Ok
D1c	...	...	65	207	312	484	Ok
D2c	...	...	309	450	64	207	Ok
D3c	...	...	244	450	248	484	Ok
Max	...	...	0.686		0.645		

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	283.843	290.000	Ok
Tube thickness :	3.376	3.404	Ok
Tube-tubesheet fillet weld leg :	2.042	3.400	Ok
Tube-tubesheet groove weld leg :	2.042	3.400	Ok

Given Shell Thickness: 125.0000 mm.
 Given Channel Thickness: 95.0000 mm.

Min Shell length of thk, (125.000) adj. to tubesheet: 806.241 mm.

Min Channel length of thk, (95.000) adj. to tubesheet: 702.865 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	204.261	1.000	199.883	1.000
Tubesheet Shear Stress	355.968	1.000	355.968	1.000
Shell Stress (Axial, Junction)	355.968	0.755	193.232	0.967
Tubesheet-Channel Junction	204.261	1.000	193.232	0.967
Tube Pressure Stress	277.380	1.000	193.232	1.000
Minimum MAWP	204.261		193.232	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	211.728	1.000	207.258	1.000
Tubesheet Shear Stress	363.489	1.000	363.488	1.000
Shell Stress (Axial, Junction)	363.489	0.696	198.485	0.958
Tubesheet-Channel Junction	211.728	1.000	198.485	0.958
Tube Pressure Stress	277.380	1.000	198.485	1.000
Minimum MAPnc	211.728		198.485	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:



Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned} &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\ &= \max(290.0/4, \min(290.0, 125.0)) \\ &= 125.000 \text{ mm.} \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned} &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\ &= \min(191.0/198.48, 1) \\ &= 0.962 \end{aligned}$$

Note: This Element/Detail was specified as being Impact Tested.

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$\begin{aligned} &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\ &= \max(290.0/4, \min(290.0, 95.0)) \\ &= 95.000 \text{ mm.} \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned} &= \min(\text{pt}/\text{Tubeside MAPnc}, 1) \\ &= \min(147.0/211.73, 1) \\ &= 0.694 \end{aligned}$$

Note: This Element/Detail was specified as being Impact Tested.

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

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